Proofs and counterexamples

Open-problem research run | 14 September 2026

The full Lyons entropy-concavity conjecture remains unresolved here. This packet presents complete arguments for restricted entropy cases, four-type fixed-point uniqueness and algorithms, and counterexamples to proposed subcritical and critical tree fluctuation limits.

Actual agents launched: **54/100**. Completed: **49/100**. Recorded model: gpt-6-astra; reasoning effort: max. The count reports execution, not mathematical validation.

Proof	Exact scope
Four-type stochastic operators	All admissible coefficients, including boundary fixed points; quantitative stability and a small-leakage extension.
Averaged iteration	Global convergence for every positive weight through 0.51; explicit uniform rates.
Determinantal entropy	Two sites; coordinate and scaled cross-block slices; indefinite rank-two directions through dimension five.
Subcritical conditioned trees	One fixed offspring law with a non-Gaussian degree-count subsequence limit.
Critical infinite-variance trees	One fixed computable offspring law with a non-Gaussian degree-count subsequence limit.

Proof status and priority are separate. The accompanying reports record independent audits and exact checks. Historical novelty is not certified. No proof-assistant verification or external referee acceptance is claimed.

The complete agent reports, sources, failed approaches, code and corrections are retained in the GitHub run directory. This PDF is a readable selection of the principal proofs, not the entire research archive.

Open the research index on GitHub

Fixed-point uniqueness for four-type strictly non-Volterra quadratic stochastic operators

Yitzchak Shmalo – research note, 14 September 2026

Statement and relation to the source problem

Every strictly non-Volterra quadratic stochastic operator on four types has exactly one fixed point. The result holds without positivity of the allowed coefficients, symmetry among the types, or an assumption that the fixed point is interior.

This settles the four-type uniqueness question left open in the current project manuscript. Ganikhodzhaev and Rozikov's 2009 survey, Problem 11 asks whether the three-type theorem, which also concerns attractiveness and cycles, extends to higher dimension. The theorem below addresses its fixed-point uniqueness clause at four types. Jamilov's 2016 publisher abstract reports uniqueness for a symmetric strictly non-Volterra operator on the three-dimensional simplex. The complete hypotheses of that paper have not yet been retrieved, so the precise comparison and worldwide novelty remain provisional. The full cycle question is outside the result established here.

The proof is elementary. A second proof yields a globally convergent averaged iteration, and the same estimates give a quantitative perturbation bound. Original discrete iterates of the quadratic operator need not converge.

1. Definitions and a transfer inequality

Write

$$\Delta_3 = \{x \in \mathbb{R}^4 : x_i \geq 0,\ \textstyle\sum_i x_i = 1\}.$$

Let

$$V_k(x) = \sum_{i,j=1}^{4} P_{ij,k} x_i x_j, \qquad P_{ij,k} = P_{ji,k} \geq 0, \qquad \sum_k P_{ij,k} = 1,$$

and suppose

$$P_{ij,k} = 0 \quad \text{whenever } k \in \{i, j\}. \tag{1}$$

These are precisely the strictly non-Volterra coefficient conditions. In particular, the polynomial defining V_k does not involve its own coordinate x_k. The map preserves the simplex because its coordinates are nonnegative and $\sum_k V_k(x) = (\sum_i x_i)^2$.

Set

$$c = \frac{3 - \sqrt{5}}{2} < \frac{1}{2}. \tag{2}$$

At a fixed point,

$$x_k = V_k(x) \leq \sum_{i,j \neq k} x_i x_j = (1 - x_k)^2.$$

Solving this quadratic inequality on $[0, 1]$ gives $x_k \leq c$ for every coordinate.

For $m \in \Delta_3$, define the matrix

$$A(m)_{rj} = \sum_a P_{aj,r} m_a. \tag{3}$$

It is nonnegative, column-stochastic, and has zero diagonal. If $m = (x + y)/2$, parent symmetry gives the exact polarization identity

$$V(x) - V(y) = 2A(m)(x - y). \tag{4}$$

Indeed $x_a x_b - y_a y_b = m_a(x_b - y_b) + (x_a - y_a)m_b$, and the two resulting sums agree.

Transfer lemma. Suppose G, H partition the four types into two pairs. For $i \in G$ and $j \in H$,

$$\mathbf{1}_G^{\mathsf{T}} A(m)(e_i - e_j) \le \max(m_i, m_j). \tag{5}$$

Proof. Let k be the other member of G and ℓ the other member of H. Put $p_{ab} = \sum_{r \in G} P_{ab,r}$. Then $0 \le p_{ab} \le 1$, $p_{ab} = p_{ba}$, $p_{ki} = 0$, and $p_{\ell j} = 1$. The last two equalities use both the prohibition (1) and the fact that each part has exactly two members. Consequently

$$\begin{aligned}
\mathbf{1}_G^{\mathsf{T}} A(m)(e_i - e_j) &= m_i(p_{ii} - p_{ij}) + m_j(p_{ij} - p_{jj}) - m_k p_{kj} + m_\ell(p_{\ell i} - 1) \\
&\le m_i(1 - p_{ij}) + m_j p_{ij} \\
&\le \max(m_i, m_j).
\end{aligned}$$

No strictly positive coordinate or coefficient is needed. □

2. Uniqueness

Theorem 1. Every operator satisfying (1) has exactly one fixed point in Δ_3.

Proof. Existence follows from Brouwer's theorem. Suppose $x \ne y$ are fixed points and write $z = x - y$. A zero-sum nonzero vector has both a positive and a negative coordinate. If z_i were its only positive coordinate, then $x_j \le y_j$ for all $j \ne i$. Since V_i is a polynomial with nonnegative coefficients in those other coordinates, $V_i(x) \le V_i(y)$, contradicting $x_i > y_i$. Interchanging x, y also rules out exactly one negative coordinate. There are therefore two strictly positive and two strictly negative coordinates.

Let $G = \{i : z_i > 0\}$, $H = \{j : z_j < 0\}$, and $h = \sum_{i \in G} z_i > 0$. Define

$$\theta_{ij} = \frac{z_i(-z_j)}{h^2}, \qquad i \in G,\ j \in H.$$

Then $\theta_{ij} \ge 0$, $\sum_{ij} \theta_{ij} = 1$, and

$$z = h \sum_{i \in G, j \in H} \theta_{ij}(e_i - e_j).$$

Every coordinate of $m = (x + y)/2$ is at most c. Equations (4) and (5) give

$$\begin{aligned}
h &= \mathbf{1}_G^{\mathsf{T}}(V(x) - V(y)) \\
&= 2h \sum_{i \in G, j \in H} \theta_{ij} \mathbf{1}_G^{\mathsf{T}} A(m)(e_i - e_j) \\
&\le 2ch < h,
\end{aligned}$$

a contradiction. This proof includes boundary fixed points. □

For a separate description of the boundary, the support of any fixed point is an invariant face: for every missing output type, a nonnegative sum over positive input coordinates vanishes, forcing every corresponding coefficient to vanish. A support of size one or two is impossible under (1). A boundary fixed point thus has support of size three; invariance of that triple together with (1) forces the omitted output coordinate to vanish identically. These observations are compatible with Theorem 1 and require no nondegeneracy assumption.

3. A constructive proof and a convergent iteration

For $c \le q < 1/2$, put

$$K_q = \{x \in \Delta_3 : \max_i x_i \le q\}.$$

Let $0 < \lambda \le q$, and consider

$$T_\lambda(x) = (1-\lambda)x + \lambda V(x). \tag{6}$$

Theorem 2. The set K_q is invariant under T_λ, and

$$\|T_\lambda(x) - T_\lambda(y)\|_1 \le L(q,\lambda)\|x-y\|_1, \quad x, y \in K_q, \quad L(q,\lambda) = 1 - \lambda(1-2q) < 1. \tag{7}$$

Proof. For $s = x_i \in [0, q]$,

$$(T_\lambda x)_i \le f(s) := (1-\lambda)s + \lambda(1-s)^2.$$

The function f is convex, so its maximum on $[0, q]$ occurs at an endpoint. Now $f(0) = \lambda \le q$, while $f(q) \le q$ because $q \ge c$. This proves invariance.

For $m = (x+y)/2$, equation (4) gives

$$T_\lambda(x) - T_\lambda(y) = B(m)(x-y), \quad B(m) = (1-\lambda)I + 2\lambda A(m).$$

All columns of B have the same sum $1+\lambda$. For any such matrix, its induced ℓ^1 norm on the zero-sum subspace is

$$\max_{i,j} \frac{1}{2}\|Be_i - Be_j\|_1 = \max_{G \subseteq \{1,2,3,4\}, i, j} \mathbf{1}_G^{\mathsf{T}} B(e_i - e_j). \tag{8}$$

The upper bound in (8) follows by decomposing an arbitrary zero-sum vector into transfers between its positive and negative coordinates; equality follows by taking a vector $e_i - e_j$. The second equality is the positive-mass formula for a zero-sum vector.

It remains to bound a subset and two distinct column indices. Empty and full subsets give zero.

- If both indices lie in G, or both outside, the identity term cancels and column stochasticity of A bounds the difference by 2λ.
- If $i \notin G$, $j \in G$, the bound is $-(1-\lambda) + 2\lambda = 3\lambda - 1 \le 2\lambda$.
- If $i \in G$, $j \notin G$, and $|G| = 1$, the A-term is $-A_{ij} \le 0$. If $|G| = 3$, it is $-A_{ji} \le 0$. Both cases give at most $1 - \lambda$.
- If $i \in G$, $j \notin G$, and $|G| = 2$, the transfer lemma gives at most $1 - \lambda + 2\lambda q$.

Finally,

$$1 - \lambda + 2\lambda q - 2\lambda = 1 - \lambda(3 - 2q) \ge 1 - q(3-2q) = (1-q)(1-2q) > 0.$$

Thus (8) is at most the constant in (7). □

Banach's theorem gives a unique fixed point of T_λ in K_q. The fixed points of T_λ and V coincide, and every fixed point of V lies in K_c, providing a second proof of Theorem 1.

Corollary 3 (global rational iteration). For

$$T(x) = \frac{3}{5}x + \frac{2}{5}V(x),$$

every initial point of Δ_3 enters $K_{2/5}$ in at most three iterations. Thereafter the iteration contracts in ℓ^1 with factor $23/25$. In particular, if $x_{n+1} = T(x_n)$ and x^* is the unique fixed point, then

$$\|x_n - x^*\|_1 \le 2(23/25)^{n-3}, \qquad n \ge 3. \tag{9}$$

Proof. Here $f(s) = 2/5 - s/5 + 2s^2/5$. Convexity successively bounds the largest coordinate by

$$1 \longmapsto \frac{3}{5} \longmapsto \frac{53}{125} \longmapsto \frac{2}{5}.$$

For the last step, $f(0) = 2/5$ and $f(53/125) = 30243/78125 < 2/5$. Invariance and (7) finish the proof; the diameter of the simplex in ℓ^1 is two. □

There are also a posteriori estimates on K_q:

$$\|x - x^*\|_1 \le \frac{\|T_\lambda x - x\|_1}{1 - L(q,\lambda)} = \frac{\|V(x) - x\|_1}{1 - 2q}. \tag{10}$$

These follow by the triangle inequality and (7). At $q = 2/5$, the final residual multiplier is five. Choosing $q = \lambda = c$ yields the smaller contraction factor

$$L(c,c) = \frac{13 - 5\sqrt{5}}{2} \approx 0.909830. \tag{11}$$

This is the minimum of the stated bound over $c \le q < 1/2$, $0 < \lambda \le q$: for fixed q it is minimized at $\lambda = q$, and $1 - q + 2q^2$ is increasing for $q \ge c > 1/4$. This optimization is within the specified invariant-set argument; it does not assert that no different algorithm has a better rate.

4. Quantitative dependence on the coefficients

Corollary 4. Let P, Q be two four-type coefficient arrays satisfying (1), with unique fixed points x_P, x_Q, and define

$$\eta = \max_{i,j} \sum_k |P_{ij,k} - Q_{ij,k}|.$$

Then

$$\|x_P - x_Q\|_1 \le \min\{2, (2\sqrt{5} - 1)\eta\}. \tag{12}$$

Proof. If $x_P = x_Q$, the conclusion is immediate; assume otherwise. Both fixed points lie in K_c. For any probability vector x,

$$\|V_P(x) - V_Q(x)\|_1 \le \sum_{i,j} x_i x_j \sum_k |P_{ij,k} - Q_{ij,k}| \le \eta.$$

Applying (10) already gives the weaker bound $\eta/(1 - 2c) = (\sqrt{5} + 2)\eta$. To retain the forced zero perturbations, write $x = x_P, y = x_Q, z = x - y$, $G = \{i : z_i > 0\}$, and $h = \|z\|_1/2$. If either sign support is a singleton, interchange P, Q and x, y, if necessary, to make the positive support the singleton. The monotonicity argument of Theorem 1 bounds the P-operator group difference by zero. The coefficient contribution to this group is at most $\eta/2$, giving $\|z\|_1 \le \eta$.

In the remaining case the signs form two pairs, $G = \{i, k\}$ and $H = \{j, \ell\}$. The transfer lemma bounds the first term in

$$h = \sum_{r \in G} [V_{P,r}(x) - V_{P,r}(y)] + \sum_{r \in G} [V_{P,r}(y) - V_{Q,r}(y)]$$

by $2ch$. The group output probability of each within-pair parental combination $\{i, k\}$ and $\{j, \ell\}$ is forced to zero or one for both arrays. Their perturbations therefore vanish. Every other row has group perturbation at most $\eta/2$, so the second term is at most

$$\frac{\eta}{2}[1 - 2(y_i y_k + y_j y_\ell)].$$

Since $y_a \le c$ and its coordinates sum to one,

$$0 \le (c-y_i)(c-y_k) + (c-y_j)(c-y_\ell) = 2c^2 - c + y_iy_k + y_jy_\ell.$$

Consequently $y_iy_k + y_jy_\ell \ge c(1-2c)$, and

$$\|z\|_1 \le \left[\frac{1}{1-2c} - 2c\right]\eta = (2\sqrt{5}-1)\eta.$$

The simplex diameter gives the other bound two. This sharper estimate is due to agent013; no optimality is claimed. □

The same sign-cut proof also improves (10) for a known approximation x: if $\|x\|_\infty < 1-c$, then

$$\|x - x^*\|_1 \le \frac{\|x - V(x)\|_1}{1 - c - \|x\|_\infty}.$$

Indeed the midpoint with the true fixed point has maximum coordinate at most $(\|x\|_\infty + c)/2$, and summing the fixed-point residual over the positive support gives this bound. Agent013 gives the full argument, including zero coordinates and residual-only variants. The solver in this packet uses the simpler rational multiplier five from (10), which remains valid.

5. The determinant question

Corollary 5. At a fixed point x^*, put $J = DV(x^*) = 2A(x^*)$, and restrict J to the tangent space $E = \{z : \sum_i z_i = 0\}$. Every positive real eigenvalue ρ of $J|_E$ obeys

$$\rho \le 2\max_i x_i^* \le 3 - \sqrt{5} < 1. \tag{13}$$

In particular,

$$\det(I - J|_E) > 0. \tag{14}$$

Proof. Take a real eigenvector for $\rho > 0$. Since the ambient matrix J is nonnegative with zero diagonal, an eigenvector with a single positive coordinate is impossible: at that coordinate $(Jz)_i \le 0 < \rho z_i$. The same argument rules out a single negative coordinate. Its signs therefore form a two-pair partition. Apply the transfer decomposition used in Theorem 1, now to $Jz = \rho z$, to obtain (13).

For $0 \le s \le 1$, $I - sJ|_E$ is nonsingular: a singularity for $s > 0$ would give a positive real eigenvalue $1/s \ge 1$, contradicting (13). Its determinant is real and continuous and equals one at $s = 0$, proving (14). □

Writing $e_3(A)$ for the sum of the principal minors of order three, (14) gives the strict inequality asked in the archived manuscript:

$$3 + 4e_3(A(x^*)) - 12\det A(x^*) > 0. \tag{15}$$

For completeness, $\operatorname{tr} A = 0$, 1 is an eigenvalue, and $\det(I - A) = 1 + e_2(A) - e_3(A) + \det A = 0$. Thus $\det(I - 2A) = 1 + 4e_2(A) - 8e_3(A) + 16\det A = -3 - 4e_3(A) + 12\det A$. The radial eigenvalue of J is two, so $\det(I - J) = -\det(I - J|_E)$, which proves (15).

6. Allowing a small probability of parental offspring

The same argument proves uniqueness for a larger class of quadratic stochastic operators. For a stochastic coefficient array on four types, define

$$\delta = \max_{i,j} \sum_{k \in \{i,j\}} P_{ij,k}. \tag{16}$$

The repeated-parent case counts the single coefficient $P_{ii,i}$ once. Thus (16) bounds the total probability that the offspring belongs to either parental type.

Theorem 6. Suppose

$$0 \le \delta < 1 - \frac{\sqrt{3}}{2}.$$

Put

$$a = 1 - \delta, \qquad s = \sqrt{1 + 4a^2}, \qquad q = 1 - \frac{s-1}{2a}, \qquad \kappa = s - 2 > 0.$$

Then V has exactly one fixed point, which belongs to K_q. For every $0 < \lambda \le q$, the averaged map $T_\lambda = (1-\lambda)I + \lambda V$ preserves K_q and contracts there with factor $L = 1 - \lambda\kappa < 1$. Its iterates converge to the fixed point from every point of the simplex. More precisely,

$$\|T_\lambda^n(x) - T_\lambda^n(y)\|_1 \le \exp\!\left(\frac{2(s-1)}{L(s+1)}\right) L^n \|x - y\|_1, \quad x, y \in \Delta_3. \tag{17}$$

The threshold is a sufficient condition; its optimality is not asserted.

Proof. The total weight of parent pairs containing type i is $1 - (1 - x_i)^2$. Therefore

$$V_i(x) \le \delta + (1 - \delta)(1 - x_i)^2 =: g(x_i).$$

The strictly decreasing function $g(u) - u$ has its unique zero in $[0, 1]$ at q, so every fixed point lies in K_q. Directly from the displayed definitions,

$$c \le q < 1 - 1/\sqrt{3} < 1/2, \qquad g(q) = q.$$

The convex function $f(u) = (1 - \lambda)u + \lambda g(u)$ has $f(0) = \lambda \le q$ and $f(q) = q$, proving invariance.

The matrix $A(m)$ in (3) remains nonnegative and column-stochastic, and now $A_{ii} \le \delta$. In a two-pair cut, the proof of (5) instead uses $p_{ki} \le \delta$, $p_{\ell j} \ge 1 - \delta$, and gives

$$\begin{aligned} \mathbf{1}_G^{\mathsf{T}} A(m)(e_i - e_j) &\le m_i(1 - p_{ij}) + m_j p_{ij} + \delta(m_k + m_\ell) \\ &\le \delta + (1 - \delta) \max_r m_r. \end{aligned}$$

For singleton and three-element separating cuts the bound is δ. The other cases in (8) are unchanged. On K_q, put $\beta = \delta + aq = (3 - s)/2$. The resulting norm bound is

$$\max\{2\lambda, 1 - \lambda + 2\lambda\beta\} = 1 - \lambda(s - 2) = L < 1.$$

The second term dominates because $\beta \ge q$, $\lambda \le q < 1/2$, and the same calculation following (8) applies. Banach's theorem proves existence and uniqueness in K_q; the coordinate bound covers every possible fixed point.

For global convergence, let $q_0 = 1$, $q_{n+1} = f(q_n)$. The identity

$$1 - qs = \frac{2a(s - 2a)}{s + 1} > 0$$

shows that f is increasing on $[q,1]$, since $\lambda \le q$. Convexity gives $T_\lambda(K_{q_n}) \subseteq K_{q_{n+1}}$, and

$$0 \le q_n - q \le (1-q)\rho^n, \qquad \rho = 1 - \frac{\lambda(s+1)}{2} \in (0,1).$$

Indeed

$$f(u) - q = (u-q)\{1 - \lambda[1 + a(2-u-q)]\},$$

whose bracket on $[q,1]$ is nonnegative and at most ρ. At stage n, the norm bound from the cut argument is at most $L + 2\lambda a(1-q)\rho^n$. Multiplying these bounds and using $1 + t \le e^t$ gives (17), because

$$\frac{2\lambda a(1-q)}{L(1-\rho)} = \frac{2(s-1)}{L(s+1)}.$$

Taking y to be the fixed point proves the global convergence assertion. □

This extension was derived by the parent, its global bound was developed by agent011, and both arguments were checked independently in agent014's full report. The strict non-Volterra case is exactly $\delta = 0$.

7. The half-step endpoint also converges

Theorem 7. Suppose V is strictly non-Volterra and satisfies (1). For $T = (I+V)/2$, every initial point converges to the unique fixed point. A coefficient-uniform explicit estimate is

$$\|T^n x - x^*\|_1 \le 2\left(\frac{7967}{8192}\right)^{n-3}, \qquad n \ge 3.$$

The complete proof is in the half-step note. Its main extra ingredient is an overlap estimate for every column pair of $A(m)$ when m is a midpoint of points in $T^3(\Delta_3)$. The sharp two-step coordinate cap is $49/128$. If both tested coordinate masses are at least $225/8192$, their common parental offspring distribution supplies that much overlap. If one mass is smaller, a preimage in $T^2(\Delta_3)$ bounds offspring probabilities into that small coordinate and forces overlap on the other two coordinates. The resulting zero-sum norm of $I/2 + A(m)$ is at most $7967/8192$. Compact invariant-image contraction then proves the theorem. Agent024 independently verified the full argument and its exact constants. This sufficient rate is not claimed optimal.

Verification and scope

The elementary proof originated in agent006's report. The transfer inequality and boundary arguments received an independent line-by-line audit in agent007. The constructive proof combines the parent derivation with agent008's independent audit and improvements. Full execution records, later checks, corrections, and exact finite computations accompany this note. Finite checks support the algebra; the universal statements rest on the proofs above. This note makes no claim that the new theorem has been formally checked in Lean or refereed by a human mathematician.

The result establishes uniqueness for every strictly non-Volterra quadratic stochastic operator on four types, and complements the project's existing five-type counterexample. It does not settle Lyons's determinantal-entropy concavity conjecture, nor the higher-dimensional cycle questions for these stochastic operators. Current literature checks and any remaining retrieval limits are recorded separately; absence of a matching search result is not proof of worldwide novelty.

An explicit convergence rate for half-averaged four-type operators

Parent derivation, 14 September 2026. This note supplies a quantitative version of agent020's qualitative half-step argument. Agent024 independently verified every inequality, the preimage choice, boundary cases, and the exact rational constants; its full report and arithmetic certificate accompany this note. All operators satisfy the strict non-Volterra conditions in `FOUR_TYPE_UNIQUENESS.md`.

Let $T = (I + V)/2$, $q = 49/128$, $d = 1 - 2q = 15/64$, and

$$\varepsilon = d^2/2 = 225/8192, \qquad R = 1 - \varepsilon = 7967/8192.$$

Then

$$\|T^n x - x^*\|_1 \le 2R^{n-3} \qquad (n \ge 3),$$

where x^* is the unique fixed point and x is any simplex point. The rate is a sufficient bound; no optimality is claimed.

The two-step cap

For a coordinate k, put $s = x_k$, $v = V_k(x)$, and $y = Tx$. Then $0 \le v \le (1-s)^2$, $y_k = (s+v)/2$, and $y_i \ge x_i/2$. The nonnegative coefficient defect implies

$$(1 - y_k)^2 - V_k(y) = \sum_{i,j \ne k} (1 - P_{ij,k}) y_i y_j \ge \tfrac{1}{4}[(1-s)^2 - v].$$

It follows by expansion that

$$(T^2 x)_k \le \frac{3 + (2s-1)v + v^2}{8}.$$

This expression is convex in v. At its two allowed endpoints it equals $3/8$ and $[3 + s^2(1-s)^2]/8$, respectively. Since $s(1-s) \le 1/4$, every coordinate is at most q. This reproduces agent008's sharp cap.

The images $S_j = T^j(\Delta_3)$ are nested and invariant under T. Thus $S_3 \subset S_2 \subset K_q$. The cap K_q itself need not be invariant.

A uniform overlap bound

For $m \in \Delta_3$, define the column-stochastic matrix

$$A(m)_{ki} = \sum_a m_a P_{ai,k}.$$

For two nonnegative unit-sum columns a, b, their total-variation distance is $1 - \sum_k \min(a_k, b_k)$. We prove that every pair of columns of $A(m)$ has overlap at least ε whenever

$$m = \tfrac{1}{2}(T^3 x + T^3 y). \tag{1}$$

Fix two different column indices. If both corresponding masses are at least ε, the columns contain the common offspring distribution $P_{ij,\cdot}$, multiplied by m_j and m_i, respectively. Their overlap is at least $\min(m_i, m_j) \ge \varepsilon$.

Otherwise let r be an index in the pair with $m_r = t < \varepsilon$, and write the other index as j; the two remaining indices are k, ℓ. Because every coordinate of m is at most q, each coordinate other than r is at least

$$1 - t - 2q = d - t > \varepsilon. \tag{2}$$

In particular, there cannot be a second coordinate below ε.

Choose z from $\{T^2x, T^2y\}$ so that $(Tz)_r \le m_r = t$. Such a choice exists by (1). Both $z_r \le 2t$ and $V_r(z) \le 2t$, while $z \in K_q$. Hence every z_a with $a \ne r$ is at least $d - 2t > 0$. For any distinct $a, b \ne r$,

$$2P_{ab,r}(d-2t)^2 \le 2P_{ab,r} z_a z_b \le V_r(z) \le 2t.$$

Thus

$$P_{ab,r} \le \theta(t) := \frac{t}{(d-2t)^2}. \tag{3}$$

For the parental pair ℓ, j, the only allowed offspring are k, r; similarly for k, j. Consequently

$$A_{kj} \ge m_\ell(1-\theta(t)), \qquad A_{\ell j} \ge m_k(1-\theta(t)). \tag{4}$$

For the pair r, j, all offspring belong to $\{k, \ell\}$, so

$$A_{kr} + A_{\ell r} \ge m_j. \tag{5}$$

The function θ is increasing on $[0, \varepsilon]$, and

$$\theta(\varepsilon) = 2048/2401 < 1.$$

Put $h = (d-t)(1-\theta(t)) > 0$. Equations (2), (4), and (5), together with $\min(a,h) + \min(b,h) \ge \min(a+b,h)$, show that the two columns overlap by at least

$$\min(A_{kr}, A_{kj}) + \min(A_{\ell r}, A_{\ell j}) \ge h.$$

Both nonnegative factors defining h decrease with t. Therefore

$$h \ge (d-\varepsilon)(1-\theta(\varepsilon)) = \frac{598335}{19668992} > \frac{225}{8192} = \varepsilon. \tag{6}$$

For the last comparison the denominator on the left is $8192 \cdot 2401$, and the numerator corresponding to the right is 540225. This proves the uniform overlap claim, including $t = 0$.

From overlap to contraction

Write $\delta(A) = \max_{i,j} \frac{1}{2}\|A(e_i - e_j)\|_1$. The overlap result gives $\delta(A(m)) \le 1 - \varepsilon$ at every midpoint (1). Polarization yields

$$Tx - Ty = B((x+y)/2)(x-y), \qquad B(m) = \tfrac{1}{2}I + A(m).$$

Its columns have equal sum $3/2$. The zero-sum ℓ^1 norm is the maximum of $\mathbf{1}_G^T B(e_i - e_j)$ over all subsets G and column pairs. If the two indices lie on the same side, the identity term cancels and the bound is $\delta(A)$. If they cross against the identity sign, the bound is $-1/2 + \delta(A)$. If they cross with its sign, singleton and triple cuts give at most $1/2$, while a two-pair cut gives at most $1/2 + q$ by the transfer lemma in the main note. Therefore

$$\|B(m)|_{\sum z_i = 0}\|_{1\to 1} \le \max\{1-\varepsilon, 1/2+q\} = R < 1.$$

For $x, y \in S_3$, their midpoint has form (1), so T is an R-contraction on the compact invariant set S_3. The fixed point belongs to this set because $x^* = T^3x^*$. Iteration and the simplex diameter two prove the displayed global rate.

This result concerns $(I+V)/2$. The original V can have a period-two orbit. No conclusion of convergence for raw iteration is implied.

Global averaged convergence beyond the half step

Parent derivation, 14 September 2026. This note extends the explicit half-step contraction. It uses the operator norm proved there. Independent audit046 verified the perturbative extension and the proof for every positive weight below one half. The stronger global range through 0.51 also passed that audit. No optimality or historical-novelty claim is made.

Let V be any strictly non-Volterra quadratic stochastic operator on four types, with symmetric parental coefficients, and put

$$T_\lambda = (1-\lambda)I + \lambda V, \qquad 0 < \lambda \le \frac{251}{500}.$$

All allowed coefficients may vanish. Let p be its unique fixed point, as proved in the main note.

Theorem. Every orbit of T_λ in the closed simplex converges to p. For $1/2 \le \lambda \le 251/500$, put

$$e = \lambda - \frac{1}{2}, \qquad R = \frac{7967}{8192}.$$

Then

$$\|T_\lambda^n x - p\|_1 \le 2(R+13e)^{n-3}, \qquad n \ge 3,$$

and

$$R + 13e \le \frac{1022499}{1024000} < 1.$$

For $0 < \lambda < 1/2$, a separate explicit bound is given in Section 3. The endpoint $251/500$ is a sufficient global range, not a proposed sharp threshold.

1. Quantitative comparison of the third images

For a vector m, define the matrix

$$A(m)_{kj} = \sum_i P_{ij,k} m_i.$$

When m is a probability vector, $A(m)$ is column-stochastic. For any real vector w, its linear dependence on w gives the ambient norm bound

$$\|A(w)\|_{1\to 1} \le \|w\|_1. \tag{1}$$

Indeed, every column is a linear combination, with coefficients w_i, of probability vectors. The exact quadratic identity

$$V(x) - V(y) = 2A((x+y)/2)(x-y) \tag{2}$$

implies that V is globally 2-Lipschitz on the simplex. For $0 \le \lambda \le 1$, T_λ is consequently $(1+\lambda)$-Lipschitz there.

Write $T = T_{1/2}$. For $e \ge 0$,

$$\|T_\lambda x - Tx\|_1 = e\|V(x) - x\|_1 \le 2e.$$

Starting from the same x, the triangle inequality yields

$$\|T_\lambda^{j+1}x - T^{j+1}x\|_1 \le (1+\lambda)\|T_\lambda^j x - T^j x\|_1 + 2e.$$

Thus, uniformly over all initial points,

$$\|T_\lambda^3 x - T^3 x\|_1 \le a_3 := 2e\big[1 + (1+\lambda) + (1+\lambda)^2\big]. \tag{3}$$

For any two initial points x, y, set

$$m_\lambda = \frac{T_\lambda^3 x + T_\lambda^3 y}{2}, \qquad m_0 = \frac{T^3 x + T^3 y}{2}.$$

Equation (3) gives $\|m_\lambda - m_0\|_1 \le a_3$.

2. Contraction on the perturbed invariant image

Let $E = \{z : \sum_i z_i = 0\}$. The full column-cut argument in the half-step note proves

$$\left\| \left(\frac{1}{2} I + A(m_0) \right) |_E \right\|_{1\to 1} \le R. \tag{4}$$

This is a norm bound on every vector in E, not merely on $T^3 x - T^3 y$.

The corresponding difference matrix for T_λ is

$$B_\lambda(m_\lambda) = (1-\lambda) I + 2\lambda A(m_\lambda).$$

Exactly,

$$B_\lambda(m_\lambda) - \left(\frac{1}{2} I + A(m_0) \right) = e(2A(m_\lambda) - I) + A(m_\lambda - m_0). \tag{5}$$

Since m_λ is a probability vector, the first matrix in parentheses has ambient norm at most 3. Equations (1), (3), (4) and (5) therefore give

$$\|B_\lambda(m_\lambda)|_E\|_{1\to 1} \le R + 3e + a_3 \le R + 13e. \tag{6}$$

For the last inequality, $1 + \lambda \le 751/500$, and

$$3 + 2\left[1 + \frac{751}{500} + \left(\frac{751}{500} \right)^2 \right] = \frac{1564501}{125000} < 13.$$

Also

$$1 - R - \frac{13}{500} = \frac{1501}{1024000} > 0. \tag{7}$$

The set $S_\lambda = T_\lambda^3(\Delta_3)$ is compact, nonempty and invariant: $T_\lambda(S_\lambda) = T_\lambda^4(\Delta_3) \subseteq S_\lambda$, since $T_\lambda(\Delta_3) \subseteq \Delta_3$. Every pair of points in S_λ has representations used above. Its difference lies in E, and (2) shows that applying T_λ multiplies that difference by $B_\lambda(m_\lambda)$. Thus (6) is a contraction estimate on S_λ, even though this set need not be convex.

The fixed point p belongs to S_λ, and the simplex has diameter 2 in ℓ^1. Iterating (6) after the first three steps proves the stated error bound. There is no appeal to global convergence from eigenvalues alone.

3. Every smaller positive averaging weight

Fix $0 < \lambda < 1/2$, and put

$$q = \max\{2/5, \lambda\} < 1/2, \qquad d = \lambda(3q - q^2 - 1) > 0,$$

$$N = \left\lceil \frac{1-q}{d} \right\rceil, \qquad L = 1 - \lambda(1-2q) < 1. \tag{8}$$

The positivity of d follows from $q \geq 2/5$, since $3(2/5) - (2/5)^2 - 1 = 1/25$, and $3q - q^2 - 1$ is increasing on $[2/5, 1]$.

Strict parent exclusion gives $V_i(x) \leq (1 - x_i)^2$. Define

$$f(u) = (1 - \lambda)u + \lambda(1 - u)^2.$$

If all coordinates of x are at most $r \in [q, 1]$, convexity of f gives

$$\max_i (T_\lambda x)_i \leq \max\{f(0), f(r)\} \leq \max\{q, r - d\}. \tag{9}$$

Here $f(0) = \lambda \leq q$, while $r - f(r) = \lambda(3r - r^2 - 1) \geq d$. Inductively, every orbit has maximum coordinate at most $\max\{q, 1 - nd\}$ after n steps. Hence $T_\lambda^N(\Delta_3) \subseteq K_q$, where K_q is the capped simplex.

Theorem 2 of the main note proves that K_q is invariant and that T_λ contracts there with factor L, because $q \geq 2/5 > (3 - \sqrt{5})/2$ and $0 < \lambda \leq q < 1/2$. Therefore

$$\|T_\lambda^n x - p\|_1 \leq 2L^{n-N}, \qquad n \geq N. \tag{10}$$

Equations (6) and (10) cover all $0 < \lambda \leq 251/500$. The entrance bound (8) is intentionally simple and need not be the fastest bound for a particular weight. Unaveraged iteration can still be unstable or cyclic, as the explicit examples in the accompanying reports show.

Global convergence through averaging weight 0.51

Parent derivation, 14 September 2026. This strengthens the smaller perturbative range in QSO_AVERAGING_RANGE.md. Independent audit046 rederived the polynomial bounds, overlap estimate, all cut cases, and the below-half extension and found no mathematical correction necessary. No optimality or historical-novelty claim is made.

Throughout, V is a four-type strictly non-Volterra QSO with symmetric parental coefficients, and $T_\lambda = (1-\lambda)I + \lambda V$. All allowed coefficients may vanish. The simplex is closed. Its unique fixed point p is established in the main theorem.

Theorem. For every $0 < \lambda \le 51/100$, every orbit of T_λ converges to p. For $1/2 \le \lambda \le 51/100$, the following uniform estimate holds:

$$\|T_\lambda^n x - p\|_1 \le 2\left(\frac{2499}{2500}\right)^{n-3}, \qquad n \ge 3. \tag{1}$$

The proof uses a two-step coordinate cap and an overlap bound on the third image. The below-half interval follows from Section 3 of QSO_AVERAGING_RANGE.md.

1. A uniform two-step cap

Fix $1/2 \le \lambda \le 51/100$, and write $a = 1-\lambda$. For a coordinate i, put $s = x_i$ and $v = V_i(x)$. Then $0 \le v \le (1-s)^2$. The symmetric bilinear form associated to V_i, denoted Q_i, satisfies

$$Q_i(x, Vx) \le (1-s)(1-v), \qquad V_i(Vx) \le (1-v)^2,$$

because any parental occurrence of i contributes zero. Expanding T_λ^2 gives

$$(T_\lambda^2 x)_i \le a^2 s + \lambda a(1+a)v + 2\lambda^2 a(1-s)(1-v) + \lambda^3(1-v)^2. \tag{2}$$

For fixed s, the right side is convex in v, so its maximum on the allowed interval occurs at $v = 0$ or $v = (1-s)^2$.

At $v = 0$, (2) equals

$$\lambda^2(2-\lambda) + a(a - 2\lambda^2)s.$$

The coefficient of s is nonpositive when $\lambda \ge 1/2$. The first term is increasing on the stated interval, and is at most

$$\left(\frac{51}{100}\right)^2 \frac{149}{100} = \frac{387549}{1000000} < \frac{2}{5}. \tag{3}$$

At the other endpoint, put $u = 1-s$. The resulting polynomial is

$$F_\lambda(u) = a^2(1-u) + \lambda a(1+a)u^2 + 2\lambda^2 a u(1-u^2) + \lambda^3(1-u^2)^2. \tag{4}$$

In the Bernstein basis of degree four,

$$F_\lambda(u) = \sum_{k=0}^{4} b_k(\lambda)\binom{4}{k}u^k(1-u)^{4-k}.$$

The coefficients and uniform upper bounds are:

k	$b_k(\lambda)$	Upper bound on $[1/2, 51/100]$
0	$1 - 2\lambda + \lambda^2 + \lambda^3$	$3/8$
1	$3/4 - 3\lambda/2 + 5\lambda^2/4 + \lambda^3/2$	$752901/2000000$
2	$(3 - 4\lambda + 6\lambda^2 - \lambda^3)/6$	$795983/2000000$
3	$1/4 + \lambda/2 - \lambda^2/4 - \lambda^3/2$	$3/8$
4	$2\lambda - 3\lambda^2 + \lambda^3$	$3/8$

These bounds follow directly by differentiation: b_0, b_3, b_4 decrease and b_1, b_2 increase throughout this interval. For example $b_0' = -2 + 2\lambda + 3\lambda^2$ is at most its negative value at $51/100$, and $b_2' = (-4 + 12\lambda - 3\lambda^2)/6$ is at least its positive value at $1/2$. The other three signs follow from their displayed quadratic derivatives. Every upper bound is strictly below $2/5$.

The Bernstein weights are nonnegative and sum to one on $[0, 1]$. Therefore $F_\lambda(u) < 2/5$, uniformly in u. Together with (2)-(3), this proves

$$T_\lambda^2(\Delta_3) \subseteq K_{2/5}. \tag{5}$$

Only the image set is claimed invariant; the entire cap need not be invariant for these weights.

2. Uniform overlap of the derivative-half columns

Set

$$q = \frac{2}{5}, \qquad d = 1 - 2q = \frac{1}{5}, \qquad \varepsilon = \frac{1}{50}.$$

Take two points in $T_\lambda^3(\Delta_3)$, and let m be their midpoint. Nested images and (5) imply $m_i \le q$. Write

$$A(m)_{kj} = \sum_i P_{ij,k} m_i.$$

This matrix is column-stochastic and has zero diagonal. We prove that every pair of its columns has total overlap at least ε.

For columns indexed by i, j with $m_i, m_j \ge \varepsilon$, the common probability vector $P_{ij,\cdot}$, weighted by m_j in column i and by m_i in column j, already supplies overlap at least ε.

Suppose instead that $m_r = t < \varepsilon$. Each other mass is at least $d - t$, since the remaining two coordinates have total at most $2q$. In particular there cannot be a second small coordinate. Among the two third-image points forming m, choose one whose r-coordinate is at most t, and write it as $T_\lambda z$, with $z \in T_\lambda^2(\Delta_3)$. Nonnegativity gives

$$z_r \le \frac{t}{1 - \lambda} \le \frac{100t}{49}, \qquad V_r(z) \le \frac{t}{\lambda} \le 2t.$$

Since $z \in K_q$, each other coordinate is at least $d - 100t/49 > 0$. For distinct $b, c \ne r$, the off-diagonal parental term appears twice, so

$$P_{bc,r} \le \frac{V_r(z)}{2 z_b z_c} \le \theta(t) := \frac{t}{(d - 100t/49)^2}. \tag{6}$$

This function is increasing on the interval in question. At its right endpoint,

$$\theta(1/50) = \frac{2401}{3042} < 1. \tag{7}$$

Now test columns r, j, and let k, ℓ be the other two types. Strict exclusion and (6) give

$$A_{kj} \geq m_\ell(1-\theta(t)), \qquad A_{\ell j} \geq m_k(1-\theta(t)),$$

while the parent pair j, r gives

$$A_{kr} + A_{\ell r} \geq m_j.$$

Put $h = (d-t)(1-\theta(t))$. The sum of the overlaps on just rows k, ℓ is at least

$$\min\{h, A_{kr}\} + \min\{h, A_{\ell r}\} \geq \min\{h, A_{kr} + A_{\ell r}\} = h.$$

Here $A_{kr} + A_{\ell r} \geq m_j \geq d - t \geq h$. Finally,

$$h \geq \left(\frac{1}{5} - \frac{1}{50}\right)\left(1 - \frac{2401}{3042}\right) = \frac{641}{16900} > \frac{1}{50}. \tag{8}$$

This covers every column pair, including zero masses. Thus the Dobrushin coefficient of $A(m)$ is at most $1 - \varepsilon = 49/50$.

3. The full zero-sum norm and convergence

The difference matrix for T_λ at midpoint m is

$$B = (1-\lambda)I + 2\lambda A(m).$$

It has equal column sums $1 + \lambda$. Its induced ℓ^1 norm on the zero-sum subspace is therefore the maximum, over column pairs i, j and row subsets G, of $\mathbf{1}_G^{\mathsf{T}} B(e_i - e_j)$.

If G contains both or neither of i, j, the identity term cancels and the cut is at most $2\lambda(49/50)$. If $i \in G, j \notin G$, the strict four-type transfer inequality gives at most $1 - \lambda + 2\lambda q$ for a two-element cut. Singleton or three-element cuts give the smaller bound $1 - \lambda$, by the zero diagonal and column sums of A. In the reversed case $j \in G, i \notin G$, stochasticity gives the bound $3\lambda - 1$. Empty and full cuts vanish. These cases exhaust the subsets.

Consequently

$$\|B|_E\|_{1\to 1} \leq \max\left\{2\lambda\frac{49}{50},\ 1 - \frac{\lambda}{5},\ 3\lambda - 1\right\} \leq \frac{2499}{2500} < 1. \tag{9}$$

Indeed the last two terms are at most $9/10$ and $53/100$, respectively.

The compact set $S = T_\lambda^3(\Delta_3)$ is invariant by nesting of images. For each of its pairs the exact identity

$$T_\lambda x - T_\lambda y = B((x+y)/2)(x-y)$$

and (9) give contraction on S. The fixed point p belongs to S; its distance from any point of the simplex is at most two. This proves (1), including the upper endpoint and all boundary or coefficient degeneracies.

The proof of convergence for $0 < \lambda < 1/2$ in QSO_AVERAGING_RANGE.md uses its scalar entrance bound and the main capped contraction theorem, so no interval of positive weights is omitted. This result is global; the larger local stability range in QSO_SPECTRAL_BOUNDS.md is a different statement.

Spectral bounds for four-type strictly non-Volterra operators

This note proves strict tangent spectral bounds, a quantitative determinant bound, and local stability of averaged iteration. It is adapted from agent029, whose complete report preserves provenance, exact algebra checks and limitations. The parent independently checked the probability, boundary and spectral arguments. Historical novelty and optimal constants are not asserted.

1. Exact statements and ambient-versus-tangent distinction

Let

$$\Delta_3 = \{x \in \mathbb{R}^4_+ : \sum_i x_i = 1\}, \qquad V_k(x) = \sum_{i,j=1}^{4} P_{ij,k} x_i x_j,$$

where

$$P_{ij,k} = P_{ji,k} \geq 0, \qquad \sum_k P_{ij,k} = 1, \qquad P_{ij,k} = 0 \quad \text{if } k \in \{i,j\}. \tag{1}$$

All allowed coefficients may vanish. Let $p \in \Delta_3$ be any fixed point, and define

$$A_{kj} = \sum_i P_{ij,k} p_i, \qquad J = DV(p) = 2A, \qquad E = \{z \in \mathbb{R}^4 : \sum_i z_i = 0\}, \qquad J_E = J|_E. \tag{2}$$

Spectra below are taken over $\mathbb{C}$, with algebraic multiplicity. Put

$$c = \frac{3-\sqrt{5}}{2}, \quad k = 2c = 3-\sqrt{5}, \quad L = \sqrt{2}, \quad a = 1-k = \sqrt{5}-2, \quad b = 1+\sqrt{2}. \tag{3}$$

The ambient matrix A is column-stochastic with zero diagonal and $Ap = p$. The ambient matrix J has column sum $\mathbf{2}$, zero diagonal, and eigenvector p with eigenvalue $\mathbf{2}$. Since $\mathbf{1}^{\mathsf{T}} p = 1$,

$$\mathbb{C}^4 = \mathbb{C}p \oplus E_{\mathbb{C}}$$

is an invariant direct sum. Consequently

$$\sigma(J) = \{2\} \uplus \sigma(J_E), \qquad \operatorname{tr} J_E = -2. \tag{4}$$

The ambient eigenvalue 2 is not a simplex tangent multiplier. In particular, neither an ambient stochastic-matrix estimate nor the zero ambient trace can be inserted unchanged into the three-dimensional restriction.

Claim 1 – all-complex spectral region and local averaging. Every tangent eigenvalue satisfies

$$\boxed{|\lambda| < \sqrt{2}.} \tag{5}$$

Every real tangent eigenvalue lies in

$$\boxed{-\sqrt{2} < \lambda \leq 3-\sqrt{5}.} \tag{6}$$

If the tangent spectrum has a nonreal pair $\alpha \pm i\beta$, $\beta \neq 0$, then its remaining real eigenvalue r obeys

$$r = -2-2\alpha, \qquad -\sqrt{2} < r \leq 3-\sqrt{5},$$

and hence

$$\boxed{\frac{\sqrt{5}-5}{2} \leq \alpha < -1+\frac{\sqrt{2}}{2} < 0, \qquad \alpha^2+\beta^2 < 2.} \tag{7}$$

Thus (7) is substantially smaller than the stochastic radius-2 disk or the half-plane $\operatorname{Re}\lambda \le 3-\sqrt{5}$.

For the averaged map

$$V_\eta = (1-\eta)I + \eta V,$$

every fixed point is locally exponentially asymptotically stable relative to the affine hyperplane $\sum x_i = 1$, and therefore relative to the simplex, whenever

$$\boxed{0 < \eta \le \eta_0 := \frac{4-\sqrt{2}}{5-\sqrt{2}} = \frac{18-\sqrt{2}}{23} \approx 0.7211211494620393.} \tag{8}$$

This is a sufficient local range, including its upper endpoint. No assertion that this is the largest valid universal range is made. No conclusion about global convergence for $\eta > 1/2$ follows from this spectral argument.

Claim 2 – determinant bounded away from zero. At every fixed point,

$$\boxed{\det(I-J_E) > D := (\sqrt{5}-2)(1+\sqrt{2})(6-\sqrt{5}-\sqrt{2}) \approx 1.3391480505809678 > \frac{4}{3}.} \tag{9}$$

The strict lower bound is proved; optimality is not. Equivalently, the previously archived identity $\det(I-J_E) = 3+4e_3(A)-12\det A$ has the stronger lower bound (9), where e_3 is the third elementary symmetric function of the four ambient eigenvalues of A. Formula (9) concerns $\det(I-J_E)$, whereas $\det(I-J) = -\det(I-J_E)$.

2. Fixed-point support and the previously established real bound

Strict exclusion gives

$$p_i = V_i(p) \le (1-p_i)^2,$$

so $p_i \le c < 1/2$. Therefore the support $S = \{i : p_i > 0\}$ has at least three elements. It has either three or four elements; all boundary fixed points in the present four-type problem are thus covered by the three-element case.

For completeness, the real bound in (6) is rederived from the transfer lemma of `agents/006.md`. Suppose $\lambda > 0$ is a real tangent eigenvalue and take a nonzero real eigenvector $z \in E$. If its only positive coordinate were z_i, nonnegativity and $J_{ii} = 0$ would give $(Jz)_i \le 0$, contrary to $\lambda z_i > 0$. Applying the same argument to $-z$ excludes exactly one negative coordinate. In four coordinates, z therefore has two positive and two negative coordinates, with no zero coordinates.

Let G and H be its positive and negative pairs. For $i \in G, j \in H$, let k' and ℓ be the other members of G and H, and write $q_{uv} = \sum_{s\in G} P_{uv,s}$. Symmetry, stochasticity, and exclusion give $q_{k'i} = 0$, $q_{\ell j} = 1$, and

$$\begin{aligned} \mathbf{1}_G^{\mathsf{T}} A(e_i - e_j) &= p_i(q_{ii}-q_{ij}) + p_j(q_{ij}-q_{jj}) - p_{k'}q_{k'j} + p_\ell(q_{\ell i}-1) \\ &\le p_i(1-q_{ij}) + p_j q_{ij} \le c. \end{aligned} \tag{10}$$

Put $h = \sum_{i\in G} z_i = -\sum_{j\in H} z_j > 0$. Then

$$z = h \sum_{i\in G, j\in H} \frac{z_i(-z_j)}{h^2}(e_i - e_j), \qquad \sum_{i\in G, j\in H} \frac{z_i(-z_j)}{h^2} = 1.$$

Using $J = 2A$ and (10) gives $\lambda h \le 2ch$, hence $\lambda \le k$. Every nonpositive real eigenvalue also satisfies this upper bound. This part is an existing input, not a new claim of worker 029.

3. Independent-parent variance bound and strictness

The following elementary argument applies on the positive support of an arbitrary symmetric QSO fixed point. The special four-type exclusion rule is needed below to control the possible missing coordinates.

First suppose all coordinates of p are positive. On $\{1,2,3,4\}$, take independent parents X, Y with law p, and conditional offspring

$$\mathbb{P}(Z = k \mid X = i, Y = j) = P_{ij,k}.$$

Because $Vp = p$, the law of Z is also p. Let f be any complex function with $\mathbb{E}_p f = 0$, and put

$$g = f(Z), \qquad h = A^{\mathsf{T}} f.$$

Parent symmetry gives both conditional expectations

$$\mathbb{E}[g \mid X] = h(X), \qquad \mathbb{E}[g \mid Y] = h(Y). \tag{11}$$

Moreover $\mathbb{E}_p h = 0$, since $Ap = p$. Independence implies that $h(X)$ and $h(Y)$ are orthogonal in the complex Hilbert space L^2. Each conditional-expectation residual is orthogonal to the corresponding parent function. Expanding the square, with complex conjugates in inner products, therefore gives the exact identity

$$\boxed{\mathbb{E}\big|g - h(X) - h(Y)\big|^2 = \|f\|^2_{L^2(p)} - 2\|A^{\mathsf{T}} f\|^2_{L^2(p)} \geq 0.} \tag{12}$$

Thus on the centered subspace $H_p = \{f : \sum_i p_i f_i = 0\}$,

$$\|A^{\mathsf{T}}|_{H_p}\|_{L^2(p)\to L^2(p)} \leq 1/\sqrt{2}. \tag{13}$$

Both constants and H_p are invariant under A^{T}, and they form a direct sum. The constant eigenvalue is 1, and (13) excludes 1 from the centered block. Hence every other ambient eigenvalue ζ of A satisfies $|\zeta| \leq 1/\sqrt{2}$, with multiplicities accounted for by this invariant decomposition.

For a finite positive support, equality cannot hold for an eigenvalue. Suppose $A^{\mathsf{T}} f = \zeta f$ with nonzero centered f and $|\zeta| = 1/\sqrt{2}$. Equality in (12) forces

$$f(Z) = \zeta(f(X) + f(Y)) \quad \text{almost surely.} \tag{14}$$

Choose i with $|f_i| = M = \max_j |f_j| > 0$. The diagonal-parent event $X = Y = i$ has positive probability p_i^2. At least one possible offspring on this event has positive conditional probability, and every possible offspring lies in the support. Equation (14) would then require

$$|f(Z)| = 2|\zeta|M = \sqrt{2}M > M,$$

contradicting maximality of M. Therefore all these eigenvalues satisfy the strict inequality $|\zeta| < 1/\sqrt{2}$. Multiplying by 2 and using (4) proves (5) for interior fixed points.

Now suppose $|S| = 3$, with missing index r. Since

$$0 = p_r = V_r(p) = \sum_{i,j\in S} P_{ij,r} p_i p_j$$

is a sum of nonnegative terms with all $p_i p_j > 0$, one has $P_{ij,r} = 0$ for every $i, j \in S$. After ordering the missing coordinate last, the matrix consequently has the block form

$$A = \begin{pmatrix} B & u \\ 0 & 0 \end{pmatrix}. \tag{15}$$

The lower right entry is zero by strict exclusion. The matrix B is the column-stochastic derivative-half matrix of the restricted three-type QSO, with strictly positive stationary vector p_S. The same argument (11)–(14) applies to B. Its two non-Perron eigenvalues have modulus less than $1/\sqrt{2}$, and the fourth ambient eigenvalue in (15) is zero. Removing the ambient eigenvalue 2 from $J = 2A$ proves (5) in the boundary case as well.

This proof does not apply a degenerate weighted norm on a zero-mass coordinate. Nor does it assume that all tangent directions at a boundary point are two-sided feasible simplex directions: the full linear tangent spectrum is obtained from the block matrix (15).

Finally, if the tangent cubic has the pair $\alpha \pm i\beta$, its third eigenvalue is real and (4) gives $r + 2\alpha = -2$. Equations (5) and (6) now give (7). In particular, the suggested weaker conclusion $\alpha \le 0$ already follows from the elementary ambient stochastic bound $r \ge -2$; the variance argument strengthens it to the strictly negative bound in (7).

4. Consequence for averaged local stability, including the endpoint

The tangent multipliers of V_η are

$$\mu = 1 - \eta + \eta\lambda. \tag{16}$$

For a real eigenvalue, (6) gives

$$1 - \eta(1 + \sqrt{2}) < \mu \le 1 - \eta(\sqrt{5} - 2) < 1.$$

These multipliers are strictly inside the unit interval provided $0 < \eta \le \eta_0$, because

$$\eta_0 < \frac{2}{1 + \sqrt{2}} = 2(\sqrt{2} - 1). \tag{17}$$

For a nonreal eigenvalue, put $\alpha_0 = -1 + \sqrt{2}/2$. For $0 < \eta \le \eta_0 < 1$, equations (5) and (7) give

$$\begin{aligned} |\mu|^2 &= (1 - \eta)^2 + 2\eta(1 - \eta)\alpha + \eta^2|\lambda|^2 \\ &< (1 - \eta)^2 + 2\eta(1 - \eta)\alpha_0 + 2\eta^2 \\ &= 1 - (4 - \sqrt{2})\eta + (5 - \sqrt{2})\eta^2 \le 1. \end{aligned} \tag{18}$$

The first strict inequality remains strict at $\eta = \eta_0$. Thus every tangent multiplier is strictly inside the complex unit disk for the full closed upper-endpoint range in (8).

For clarity about the dynamical inference: a finite-dimensional linear map with spectral radius less than 1 admits an equivalent norm with operator norm less than 1. The polynomial map V_η, restricted to the invariant affine hyperplane $\sum_i x_i = 1$, then has derivative norm less than 1 in a sufficiently small neighborhood of p. This proves local exponential asymptotic stability by the mean-value inequality, including boundary fixed points when restricted back to the simplex. This argument does not require diagonalizability and does not use fixed-point uniqueness as a stability criterion.

The ambient derivative of V_η has the additional radial eigenvalue $1 + \eta > 1$. There is no claim of stability in all of $\mathbb{R}^4$.

Exact illustration: nonreal multipliers can destabilize the original map

Specify deterministic offspring for the ten unordered parent pairs by

Parent pair	Offspring
11	2
22	3
33	1
12	3
13	2
23	1
14	2
24	3
34	1
44	1

Take the same offspring after interchanging parents. This is an admissible four-type operator. The face $x_4 = 0$ is invariant and contains the fixed point $p = (1/3, 1/3, 1/3, 0)$. At this point,

$$J = \begin{pmatrix} 0 & 2/3 & 4/3 & 2/3 \\ 4/3 & 0 & 2/3 & 2/3 \\ 2/3 & 4/3 & 0 & 2/3 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \sigma(J_E) = \left\{0, -1 + \frac{i}{\sqrt{3}}, -1 - \frac{i}{\sqrt{3}}\right\}. \tag{19}$$

The pair has modulus $2/\sqrt{3} > 1$. On the two-dimensional invariant face, the fixed point is a hyperbolic repeller; hence it is unstable relative to the full simplex for the original unaveraged map. This illustrates the precise limitation of any inference from uniqueness, the continuous flow, or stable sufficiently averaged iteration. It is not an extremal example for (5), (8), or (9), and no example-optimality claim is made.

5. Proof of the quantitative determinant bound

All real tangent eigenvalues are less than 1, and every nonreal pair contributes a positive squared modulus to $\det(I - J_E)$. There are two cases.

5.1. Three real eigenvalues

Let $t_i = 1 - \lambda_i$. Equations (4)–(6) imply

$$a \le t_i < b, \qquad t_1 + t_2 + t_3 = 5, \qquad \det(I - J_E) = t_1 t_2 t_3. \tag{20}$$

Consider the closed polytope obtained by replacing $t_i < b$ with $t_i \le b$. Its positive product is minimized at a vertex: equivalently, $\sum_i \log t_i$ is strictly concave, so a nonvertex point cannot be a minimum. Every vertex has two coordinates at interval endpoints. The possibilities with two lower endpoints or two upper endpoints are infeasible, because

$$2a + b < 5 < a + 2b. \tag{21}$$

Thus the vertices are exactly the permutations of

$$(a, b, 5 - a - b) = (\sqrt{5} - 2, 1 + \sqrt{2}, 6 - \sqrt{5} - \sqrt{2}).$$

Their product is D. Every such vertex has a coordinate equal to b, which is excluded by the strict spectral bound in (20). Therefore the actual determinant is strictly larger than D.

5.2. One real eigenvalue and a nonreal pair

Let the real eigenvalue be r. By the trace identity, $\alpha = -1 - r/2$, and therefore

$$\begin{aligned} \det(I - J_E) &= (1-r)((1-\alpha)^2 + \beta^2) \\ &> (1-r)\frac{(r+4)^2}{4} =: f(r). \end{aligned} \tag{22}$$

Here $1 - r > 0$ and $\beta^2 > 0$. On $[-\sqrt{2}, k]$,

$$f'(r) = -\frac{3}{4}(r+4)(r+2) < 0.$$

Since $-\sqrt{2} < r \le k$, one gets

$$f(r) \ge f(k) = \frac{41\sqrt{5} - 89}{2}.$$

Writing $a = 1 - k$ and $b = 1 + \sqrt{2}$ as in (3), the exact identity

$$f(k) - D = a\left(b - \frac{5-a}{2}\right)^2 \ge 0 \tag{23}$$

proves the same strict bound in this case.

Finally,

$$D = 6\sqrt{5} + 7\sqrt{10} - 15\sqrt{2} - 13.$$

For a short rational certificate of $D > 4/3$, use $2.236 < \sqrt{5} < 2.237$ and $1.414 < \sqrt{2} < 1.415$, verified simply by squaring their positive rational endpoints. The factored form of D gives

$$D > \frac{59}{250}\frac{1207}{500}\frac{587}{250} = \frac{41802031}{31250000} > \frac{4}{3}. \tag{24}$$

Equations (20)–(24) prove Claim 2 in all cases, including repeated real eigenvalues, boundary fixed points, and nondiagonalizable tangent matrices.

Verification and scope

The exact script 029_spectral_exact.py checks radical identities, rational bounds, and the example characteristic polynomial. The universal statements follow from the proofs above. The averaging range beyond one half is a local conclusion; this note proves no global convergence there. No proof-assistant verification or worldwide-priority claim is made.

Verified restricted cases of determinantal entropy concavity

Research note, 14 September 2026. The general conjecture remains unresolved in this run. The results below are session derivations with proofs and recorded audits; historical novelty is not established.

For a complex Hermitian positive contraction $0 \le K \le I$ on a finite coordinate set E, let P_K be the determinantal law characterized by

$$P_K(S \subseteq X) = \det K_S,$$

and let $H(K)$ be its Shannon entropy, using natural logarithms. Lyons's Conjecture 2.6 asks whether H is concave on the whole contraction set. Gu's rank-one result is established prior work.

The following statements have complete proofs in this packet:

1. Full concavity for two sites, with the sharp interior bound $D^2H(K)[D,D] \le -8\det D$ when $\det D > 0$.
2. In any dimension, concavity when the varying entries lie in a fixed two-coordinate principal block and all other entries remain fixed.
3. In any dimension, concavity when the principal block complementary to one coordinate remains fixed; its incident row and column may vary together.
4. Through dimension five, concavity along every direction with at most one positive and at most one negative eigenvalue.
5. In any dimension, every fixed nonzero Hermitian line through an interior diagonal kernel is strictly concave on a sufficiently small interval around that kernel. This is a local statement along each fixed line.

These are different restrictions. None allows arbitrary sums of its permitted directions without a separate control of mixed Hessian terms. In particular, the fourth statement does not include positive semidefinite directions of rank two.

1. Two sites: the full theorem

Write

$$K = \begin{pmatrix} a & c \\ \bar{c} & b \end{pmatrix}, \qquad D = \begin{pmatrix} u & v \\ \bar{v} & s \end{pmatrix}, \qquad 0 < K < I.$$

The exact atom probabilities are

$$x = (1-a)(1-b) - |c|^2, \quad y = a(1-b) + |c|^2, \quad z = (1-a)b + |c|^2, \quad w = ab - |c|^2.$$

They are positive, sum to one, and satisfy $yz - xw = |c|^2$. With $\delta = \det D$, direct differentiation along $K + tD$ gives

$$(x'', y'', z'', w'') = 2\delta(1, -1, -1, 1),$$

and hence

$$H'' = -\mathcal{I} + 2\delta L, \qquad \mathcal{I} = \sum_{p \in \{x,y,z,w\}} \frac{(p')^2}{p}, \qquad L = \log \frac{yz}{xw} \ge 0. \tag{1}$$

This already proves $H'' \le 0$ when $\delta \le 0$.

If $\delta > 0$, the direction is definite. Negation preserves its second variation, so take $D > 0$. Set

$$A = \frac{\operatorname{adj} K}{\sqrt{w}}, \qquad B = \frac{\operatorname{adj}(I-K)}{\sqrt{x}}, \qquad T = \frac{y+z}{2\sqrt{xw}} \ge 1.$$

Both A, B are positive definite of determinant one. The two extreme Fisher terms alone give

$$\begin{aligned}\mathcal{I} &\geq (\operatorname{tr} AD)^2 + (\operatorname{tr} BD)^2 \\ &\geq \tfrac{1}{2}[\operatorname{tr}(A+B)D]^2 \geq 2\det(A+B)\det D.\end{aligned} \tag{2}$$

The last step applies arithmetic-geometric mean to the two eigenvalues of $(A+B)^{1/2}D(A+B)^{1/2}$, and needs no commutativity. The two-dimensional determinant identity yields

$$\det(A+B) = 2 + \frac{y+z}{\sqrt{xw}} = 2 + 2T.$$

Also $L \leq 2\log T$. Substituting into (1),

$$H'' \leq -4\delta(1 + T - \log T) \leq -8\delta. \tag{3}$$

Equality is attained at $K = I_2/2, D = I_2$; the coefficient eight is sharp. The exact minimum of the two extreme Fisher terms over $D > 0, \det D = 1$ is $2\det(A+B)$, attained at $D = \sqrt{\det(A+B)}(A+B)^{-1}$. No minimum of the full four-atom Fisher expression is asserted by this calculation.

Integrating the nonpositive Hessian proves concavity on strict segments. For closed contractions, use $K_j^\epsilon = (1-2\epsilon)K_j + \epsilon I$ and pass to the limit. Each atom is polynomial in the entries, and $-p\log p$ is continuous at zero. When $\det(K_1 - K_0) > 0$, (3) also gives

$$H((1-t)K_0 + tK_1) \geq (1-t)H(K_0) + tH(K_1) + 4\det(K_1 - K_0)t(1-t). \tag{4}$$

The complete derivation and independent audit, including equality and complex phases, are in agent016 and agent018.

2. A fixed two-coordinate block in any dimension

Write the kernel in blocks for $E = U \sqcup O$, where $|U| = 2$, and suppose only K_{UU} varies. For an outside atom $Y \subseteq O$, put

$$Q_Y = K_{OO} - I_{O\setminus Y}, \qquad q_Y = (-1)^{|O\setminus Y|}\det Q_Y.$$

The numbers q_Y are the fixed outside probabilities. If $q_Y > 0$, the conditional law on U is determinantal with kernel

$$C_Y = K_{UU} - K_{UO}Q_Y^{-1}K_{OU}. \tag{5}$$

Indeed inclusion-exclusion in the absent outside coordinates, followed by the Schur complement, gives

$$P(W \subseteq X,\ X \cap O = Y) = q_Y \det(C_Y)_{WW}, \qquad W \subseteq U.$$

The conditional kernel is a positive contraction: this follows by conditioning on included outside coordinates first, using a positive Schur complement, then excluding the others by the complementary Schur complement. Continuity covers boundary kernels with $q_Y > 0$; zero-weight atoms are omitted.

For $K_{UU}(t) = K_{UU}(0) + td$, formula (5) gives $C_Y(t) = C_Y(0) + td$ with the same direction in every conditional law. The entropy chain rule is

$$H(K(t)) = H((q_Y)_Y) + \sum_{Y:q_Y>0} q_Y H(C_Y(t)).$$

The first term and weights are fixed. The two-site theorem proves concavity, including boundary segments. If $\det d > 0$, the chord improvement is $4\det d\, t(1-t)$, since the weights sum to one. Agent018 gives the full conditional-kernel derivation.

3. Fixing the complementary block of one coordinate

For a real set function f, let $\widehat{f}$ denote its Boolean Möbius coefficients. Its determinantal expectation is the polynomial

$$F_f(K) = \mathbb{E}_K f(X) = \sum_{S \subseteq E} \widehat{f}(S) \det K_S. \tag{6}$$

Choose an original coordinate i, write $R = E \setminus \{i\}$, and set

$$K = \begin{pmatrix} a & b^* \\ b & C \end{pmatrix}, \quad f_0(T) = f(T), \quad g(T) = f(T \cup \{i\}) - f(T).$$

Bordered determinants in (6) give the exact identity

$$F_f(K) = F_{f_0}(C) + aF_g(C) - b^* Q_g(C) b, \quad Q_g(C) = \nabla F_g(C). \tag{7}$$

The gradient is with respect to the real trace pairing on Hermitian matrices. If g is increasing, then $Q_g(C) \geq 0$ for $0 < C < I$. This follows from the established determinantal stochastic-domination theorem: increasing a positive-contraction kernel in Loewner order increases the expectation of every increasing set function. Applying it to $C + sww^*$ and differentiating gives $w^* Q_g(C) w \geq 0$. This input is Lyons's Theorem 2.9.

At an interior base kernel freeze $f(S) = -\log p_S(K)$, where p_S are exact atom probabilities. This function is supermodular. To check it, put $L = K(I-K)^{-1} > 0$, so $p_S = \det(I-K)\det L_S$; the two-by-two Schur complement gives

$$p_{T\cup\{i\}} p_{T\cup\{j\}} \geq p_T p_{T\cup\{i,j\}}.$$

Thus $g = \Delta_i f$ is increasing. Along a direction

$$D = \begin{pmatrix} \alpha & v^* \\ v & 0 \end{pmatrix},$$

with C fixed, differentiating (7) twice yields $D^2 F_f(K)[D,D] = -2v^* Q_g(C) v$. The entropy Hessian is therefore

$$D^2 H(K)[D,D] = -\sum_S \frac{(Dp_S[D])^2}{p_S} - 2v^* Q_g(C) v \leq 0. \tag{8}$$

The table f is frozen during this auxiliary differentiation. Integration and the same common regularization prove the stated slice concavity on the closed contraction set. Agent010 contains the derivation; agent017 independently reconstructs it both by Möbius expansion and conditional binary entropy.

4. Arbitrarily oriented indefinite rank-two directions through dimension five

For supermodular f, define

$$q_f(K; u, v) = D^2 F_f(K)[uu^*, vv^*].$$

We show $q_f \geq 0$ when $|E| \leq 5$. Dependent vectors give zero; otherwise orthonormalizing their plane scales q_f by the nonnegative Gram determinant. Hence take u, v orthonormal and put $U = \operatorname{span}(u, v)$, $R = P_{U^\perp}$.

The mixed coefficient of each principal determinant depends only on the compression of K orthogonal to the restricted vectors u_S, v_S. Extending such perpendicular vectors by zero puts them in $U^\perp$. Thus

$$q_f(K; u, v) = q_f(RKR; u, v). \tag{9}$$

Diagonalize this compression as $\sum_{j=1}^{n-2} \lambda_j w_j w_j^*$. Separate affinity in the λ_j expresses (9) as a convex combination of values at projections onto spans of subsets of the w_j.

For a rank-zero compressed projection, (6) gives

$$q_f(0; u, v) = \mathbb{E}_{Y \sim P_U} \widehat{f}(Y) \geq 0,$$

because $|Y| = 2$ and $\widehat{f}(\{i, j\}) = \Delta_i \Delta_j f(\varnothing)$. For a rank-one projection onto a unit vector $w \perp U$, Cauchy-Binet gives

$$q_f(ww^*; u, v) = \mathbb{E}_{Y \sim P_U} \widehat{f}(Y) + \mathbb{E}_{Z \sim P_{U \oplus [w]}} \widehat{f}(Z). \tag{10}$$

Established stochastic domination for nested projection subspaces couples $Y \subset Z$. Their sizes are two and three. Writing $Y = \{i, j\}, Z = \{i, j, k\}$, their coupled sum of coefficients is

$$\widehat{f}(ij) + \widehat{f}(ijk) = \Delta_i \Delta_j f(\{k\}) \geq 0.$$

This proves (10) nonnegative in every ambient dimension.

Complementation sends f to the supermodular function $f^c(S) = f(E \setminus S)$ and gives $F_f(K) = F_{f^c}(I - K)$. The two derivative signs cancel. By (9), a compressed projection of rank r may therefore be replaced by one of rank $n - 2 - r$. For $n \leq 5$, every projection term reduces to rank zero or one. Hence $q_f \geq 0$.

Finally freeze the supermodular $f = -\log p_K$ as before. For $D = \lambda uu^* - \mu vv^*$, with $\lambda, \mu \geq 0$, pure rank-one second derivatives vanish and

$$D^2 H(K)[D, D] = -\sum_S \frac{(Dp_S[D])^2}{p_S} - 2\lambda\mu q_f(K; u, v) \leq 0.$$

Integration and common regularization prove the finite-segment claim, including the boundary. Agent019 gives the full compression proof, exact rational identity checks, and the unresolved higher-dimensional obligation.

5. Exact quartic loss at an interior diagonal kernel

Let $D = \operatorname{diag}(p_i)$, with $0 < p_i < 1$, and let $A \neq 0$ be a Hermitian matrix with zero diagonal. Put $v_i = p_i(1 - p_i)$ and $h(p) = -p \log p - (1 - p) \log(1 - p)$. Then

$$H(D + tA) = \sum_i h(p_i) - \frac{t^4}{2} \sum_{i<j} \frac{|A_{ij}|^4}{v_i v_j} + O(t^6). \tag{11}$$

This implies strict concavity on a sufficiently short interval along this fixed line, even though its second derivative at zero vanishes.

Here is the full argument for the leading term. Let $\mu = \bigotimes_i \text{Bernoulli}(p_i)$, and define the centered contrasts $z_i = (X_i - p_i)/v_i$, $z_J = \prod_{i \in J} z_i$. The mixed-column determinant formula gives the exact density polynomial

$$\frac{P_{D+tA}(X)}{\mu(X)} = 1 + \sum_{\varnothing \neq J} t^{|J|} \det(A_J) z_J =: 1 + u_t.$$

The zero diagonal eliminates its singleton terms. Product independence gives $\mathbb{E}_\mu z_J z_L = 0$ for $J \neq L$, and $\mathbb{E}_\mu z_J^2 = \prod_{i \in J} v_i^{-1}$. The marginals are unchanged, so the cross-entropy against μ is constant and

$$H(D + tA) = H(D) - \mathbb{E}_\mu[(1 + u_t) \log(1 + u_t)].$$

There are finitely many positive atoms near zero, so this expression is analytic. Write $u_t = t^2F_2 + t^3F_3 + \cdots$, with

$$F_2 = -\sum_{i<j} |A_{ij}|^2 z_i z_j.$$

The expansion $(1+u)\log(1+u) = u + u^2/2 - u^3/6 + \cdots$, normalization $\mathbb{E}u_t = 0$, and orthogonality give

$$\mathbb{E}[(1+u_t)\log(1+u_t)] = \frac{t^4}{2}\mathbb{E}F_2^2 + O(t^6) = \frac{t^4}{2}\sum_{i<j}\frac{|A_{ij}|^4}{v_i v_j} + O(t^6).$$

The fifth coefficient vanishes because $\mathbb{E}F_2F_3 = 0$; every cubic power of u_t has order at least six. This proves (11). The coefficient sum is strictly positive, and analytic differentiation gives $H''(t) = -6t^2\sum_{i<j}|A_{ij}|^4/(v_iv_j) + O(t^4) < 0$ for sufficiently small nonzero t. Integrating the derivative proves strict concavity across zero as well.

For a general Hermitian direction with a nonzero diagonal entry, the same density calculation instead yields $H''(0) = -\sum_i A_{ii}^2/v_i < 0$, so local strict concavity follows by continuity. Thus every fixed nonzero Hermitian line is covered. Agent023 contains the all-order formula, independently checked finite polynomial identities, and examples showing that later coefficients need not all be negative or even. The parent independently checked the product-contrast proof. This controls the derivative along the line from D; it does not prove concavity for every independent direction at every nearby kernel.

6. A scaled cross matrix in any dimension

Fix two actual point coordinates U, their complement O, a Hermitian outside contraction C, and an arbitrary complex $2 \times |O|$ matrix B. Then

$$\mathcal{H}(A,z) = H(P_{K(A,z)}), \qquad K(A,z) = \begin{pmatrix} A & zB \\ \bar{z}B^* & C \end{pmatrix} \tag{12}$$

is jointly concave on the closed convex domain where $A = A^*$, $z \in \mathbb{C}$, and $0 \le K(A,z) \le I$. The entire cross matrix shares one scalar; its entries cannot vary independently.

Here is a finite-chord proof, including a quantitative strictness statement. Initially assume the endpoint kernels are strict contractions. For an outside atom $Y \subset O$, put

$$Q_Y = C - P_{O\setminus Y}, \quad q_Y = (-1)^{|O\setminus Y|}\det Q_Y > 0, \quad D_Y = BQ_Y^{-1}B^*.$$

The q_Y are the fixed outside probabilities. Inclusion-exclusion on absent outside sites and a Schur complement give the conditional inside kernel

$$M_Y(A,z) = A - |z|^2 D_Y. \tag{13}$$

Each is a strict contraction. Indeed, conditioning first on included sites gives a positive Schur complement T with $T \le K < I$. Conditioning next on excluded sites gives $T_{UU} + T_{UZ}(I - T_{ZZ})^{-1}T_{ZU}$; its complement is a Schur complement of $I - T > 0$. This also proves the assertion for empty included or excluded sets. No fixed sign of D_Y is assumed.

For $s \ge 0$, the full kernel $K(A, \sqrt{s})$ is strict exactly when

$$A - sBC^{-1}B^* > 0, \qquad I - A - sB(I-C)^{-1}B^* > 0. \tag{14}$$

Hence the feasible (A, s) domain is convex, and at fixed A it contains the interval from zero to any feasible s. By the two-site theorem, the function

$$F_A(s) = \sum_Y q_Y H_2(A - sD_Y)$$

is concave in s on that interval. The chain rule gives $\mathcal{H}(A, z) = H(P_C) + F_A(|z|^2)$, and $F_A(0) = H_2(A)$.

For endpoints (A_j, z_j) and $0 < t < 1$, set

$$\begin{aligned} A_t &= (1-t)A_0 + tA_1, & z_t &= (1-t)z_0 + tz_1, \\ \bar{s} &= (1-t)|z_0|^2 + t|z_1|^2, & \Delta &= t(1-t)|z_1 - z_0|^2. \end{aligned}$$

Equation (14) proves feasibility of $\bar{K} = K(A_t, \sqrt{\bar{s}})$. Its conditional kernels are convex combinations of the endpoint conditional kernels. The two-site theorem therefore gives

$$H(P_{\bar{K}}) \geq (1-t)\mathcal{H}(A_0, z_0) + t\mathcal{H}(A_1, z_1). \tag{15}$$

If $\bar{s} > 0$, write $\alpha = \Delta/\bar{s} \in [0, 1]$. Since $|z_t|^2 = (1-\alpha)\bar{s}$, concavity of F_{A_t} between zero and $\bar{s}$ gives

$$F_{A_t}(|z_t|^2) - F_{A_t}(\bar{s}) \geq \alpha\,[H_2(A_t) - F_{A_t}(\bar{s})] = \alpha I_{P_{\bar{K}}}(X \cap U; X \cap O).$$

Combining this with (15) proves

$$\mathcal{H}(A_t, z_t) - (1-t)\mathcal{H}(A_0, z_0) - t\mathcal{H}(A_1, z_1) \geq \frac{\Delta}{\bar{s}} I_{P_{\bar{K}}}(X \cap U; X \cap O) \geq 0. \tag{16}$$

If $\bar{s} = 0$, both scalars vanish, so ordinary two-site concavity applies.

For closed endpoints, replace every kernel by $(1 - 2\varepsilon)K + \varepsilon I$. This keeps one fixed outside block and one fixed cross matrix, with scalars multiplied by $1 - 2\varepsilon$. The ratio $\Delta/\bar{s}$ is unchanged, and the auxiliary kernel is regularized in the same way. Letting $\varepsilon \downarrow 0$ proves its feasibility and (16) by continuity of finite-law entropy and mutual information. Inverses are used only in the strict proof.

If $B \neq 0$ and $z_0 \neq z_1$, some cross entry satisfies

$$\mathrm{Cov}_{P_{\bar{K}}}(\mathbf{1}_{i \in X}, \mathbf{1}_{j \in X}) = -\bar{s}|B_{ij}|^2 < 0.$$

The two subsets are dependent, so their mutual information is positive. Thus every interior point of the chord has strict entropy inequality, including for boundary endpoints. The argument was developed by coordinator011 from agent036's three-site case; agent045 independently audited the all-dimensional proof, supplied (16), and checked the exact conditional identities on a genuinely complex five-site example. The parent independently reviewed that full report. The same proof lifts any proved k-site concavity theorem to a k-coordinate block, but only $k = 2$ is used unconditionally here.

What the proofs do not establish

The general Lyons conjecture is unresolved here, including unrestricted directions in dimensions three and above. The supermodular mixed-derivative lemma above first leaves an unresolved compressed rank-two projection case in dimension six. The bounded numerical search in `computations/dpp_supermodular_projection_search.py` found no counterexample among its 100 sampled real configurations; this is not a proof of that lemma. Positivity on decomposable exterior two-vectors also does not establish positivity of the full exterior-square operator.

Arbitrary unitary rotation is not a symmetry of the entropy problem. For example, $\mathrm{diag}(1,0)$ has entropy zero, while the unitarily conjugate projection with every entry $1/2$ has entropy $\log 2$. Coordinate restrictions above cannot be removed by rotating the kernel.

No theorem in this note is claimed to have been formally checked in Lean, externally refereed, or cleared for worldwide priority. Exact proofs, audit disagreements and their resolutions, and current source limitations are preserved in the accompanying reports and final claim ledger.

A fixed subcritical offspring law with non-Gaussian degree-count fluctuations

Research proof assembled in this run, 14 September 2026. This is a complete repair of the archived rare-pair construction, with a fixed explicit law and a non-Gaussian subsequence limit. The source match is Thévenin's subcritical closing conjecture. A separate critical infinite-variance construction addresses the other clause. Historical novelty and current literature priority are not certified.

The derivation is adapted from agent015, with the parent independently checking the construction, conditioning estimates, Fourier lemma, and tree transfer. The full independent audit 034 found no mathematical gap and added the moment threshold, joint-lattice clarification and cyclic endpoint detail incorporated below.

1. Exact theorem

There is a probability law `pi` on the nonnegative integers with

$$\pi_1 = q = \frac{1}{4}, \qquad \pi_0 > \frac{1}{2}, \qquad \frac{1}{4} < \nu := \mathbb{E}\xi < \frac{5}{16}, \qquad 0 < \sigma^2 := \operatorname{Var}\xi < \infty,$$

and `E exp(theta xi)=infinity` for every `theta>0`, such that the following holds. Let `T_n` be the Galton-Watson tree with offspring law `pi`, conditioned to have exactly `n` vertices; it is defined for every positive integer `n`. Put

$$n_r = 2^{8\cdot 4^{r-1}}, \qquad \gamma = q(1-\nu), \qquad a_* := \frac{\gamma}{\sigma^2}, \qquad v_* := q(1-q) - \frac{\gamma^2}{\sigma^2}. \tag{1}$$

Then `a_*>0`, `v_*>0`, and

$$\frac{N_1(T_{n_r}) - n_r/4}{\sqrt{n_r}} \implies \frac{1}{2}\mathcal{N}(-a_*, v_*) + \frac{1}{2}\mathcal{N}(a_*, v_*). \tag{2}$$

The law on the right is not Gaussian. In particular, the full sequence cannot converge to `N(0,3/16)` under this centering and scaling. Along this subsequence the tree also has one degree asymptotic to `(1-nu)n_r`, while its second-largest degree is `o(n_r)`, with probability tending to one.

2. One fixed law, including exact integer support

Set

$$n_1 = 256, \quad n_{r+1} = n_r^4, \quad p_r = n_r^{-4}, \quad d_0 = \frac{3}{4}.$$

Recursively define

$$H_r = d_{r-1}n_r, \qquad L_r^{\pm} = H_r \pm \sqrt{n_r}, \qquad d_r = d_{r-1}(1 - 2n_r^{-3}). \tag{3}$$

Assign masses

$$\pi_1 = \frac{1}{4}, \qquad \pi_{L_r^-} = \pi_{L_r^+} = p_r, \qquad \pi_0 = \frac{3}{4} - 2\sum_{r\ge 1} p_r, \tag{4}$$

with all remaining masses zero. These formulas define the law once and for all. Neither the probabilities nor the support are adjusted after a conditioning size is chosen.

For `a>0` and every `r`, scale growth gives the useful bound

$$\sum_{s\ge r} n_s^{-a} \le \sum_{j\ge 0} n_r^{-a(j+1)} = \frac{n_r^{-a}}{1-n_r^{-a}}. \tag{5}$$

For the exponents `a=2,3,4` used below, this is at most `2n_r^{-a}`.

The construction has integer support without rounding. To see this inductively, write

$$c_r := \sum_{s\le r} p_s(L_s^- + L_s^+) = \frac{3}{4} - d_r. \tag{6}$$

The identity follows from `p_r(L_r^-+L_r^+)=2d_{r-1}n_r^{-3}`. Assuming all earlier atoms are integers, each denominator `n_s^4`, for `s<r`, divides `n_r`; moreover `n_r/4` is an integer. Therefore

$$H_r = \Big(\frac{3}{4} - c_{r-1}\Big) n_r$$

is an integer. Also `sqrt(n_r)` is an integer. The base case is `H_1=192`, giving `L_1^-=176` and `L_1^+=208`.

Every `d_r` is positive and at most `3/4`. From (5) and (6),

$$0 < c_r \le \frac{3}{2}\sum_{s\ge 1} n_s^{-3} \le 3\cdot 256^{-3} < \frac{1}{16}.$$

Consequently,

$$\frac{11}{16} < d_r \le \frac{3}{4}, \qquad \frac{5}{8} n_r < L_r^- < L_r^+ \le \frac{13}{16} n_r < n_r. \tag{7}$$

Here `sqrt(n_r)/n_r<=1/16` was used. Further, `L_{r+1}^->(5/8)n_r^4>n_r`. Thus different stages do not collide, no rare atom is zero or one, and every earlier-stage atom is below `n_{r-1}`.

The total mass assigned to rare atoms is at most `4n_1^{-4}<1/4`; hence (4) is a probability law with `pi_0>1/2`. Its mean is

$$\nu = q + \sum_{r\ge 1} p_r(L_r^- + L_r^+) = 1 - d_\infty, \qquad d_\infty = \frac{3}{4}\prod_{r\ge 1}(1-2n_r^{-3}) \in \Big(\frac{11}{16}, \frac{3}{4}\Big). \tag{8}$$

Its second moment satisfies

$$\mathbb{E}\xi^2 = q + \sum_{r\ge1} p_r\big((L_r^-)^2 + (L_r^+)^2\big) \le \frac{1}{4} + 2\sum_{r\ge1} n_r^{-2} \le \frac{1}{4} + 4\cdot 256^{-2} < \frac{1}{2}. \tag{9}$$

In particular, its variance is finite. It is positive, even at least `pi_0 pi_1>1/8`, by the independent-copy identity for variance and the atoms at zero and one. The support has lattice span one.

The full independent audit 034 also gives the exact moment threshold:

$$\mathbb{E}\xi^h < \infty \quad (0 < h < 4), \qquad \mathbb{E}\xi^h = \infty \quad (h \ge 4).$$

For the first assertion the rare contribution is at most $2\sum_r n_r^{h-4}$. For the second, a single atom at each stage contributes at least $(5/8)^h n_r^{h-4}$. Thus this subcritical counterexample has a finite third moment.

For every `theta>0`, individual terms of the exponential-moment sum satisfy

$$p_r e^{\theta L_r^-} \ge n_r^{-4} e^{5\theta n_r/8} \longrightarrow \infty. \tag{10}$$

Thus there is no positive exponential moment. The probability generating function has radius exactly one. If the simply generated weights are chosen to be `w_k=pi_k`, their canonical law in Janson's subcritical parametrization is this same `pi`: `rho=tau=1`, `Phi(1)=1`, and `Psi(1)=nu<1`.

3. The correct truncated law and its uniform bounds

For a fixed stage `r`, abbreviate `n=n_r`, and let

$$\mathcal{A}_r = \{0,1\} \cup \{L_s^-, L_s^+ : s < r\}, \qquad t_r = 2\sum_{s\ge r} p_s, \qquad z_r = 1 - t_r.$$

Let `X_r` have the law of `xi` conditioned on membership in `A_r`. Write

$$u_r = q + c_{r-1} = 1 - d_{r-1}, \qquad \mu_r = \mathbb{E}X_r = \frac{u_r}{z_r}, \qquad q_r = \mathbb{P}(X_r = 1) = \frac{q}{z_r}. \tag{11}$$

By (5),

$$0 < t_r \le 4n^{-4}, \qquad z_r > \frac{3}{4}, \qquad \mu_r - u_r = \frac{u_r t_r}{z_r} = O(n^{-4}), \qquad q_r - q = \frac{q t_r}{z_r} = O(n^{-4}). \tag{12}$$

The implicit constants here are numerical, independent of the stage. Also

$$\mu_r < \frac{5/16}{3/4} = \frac{5}{12} < \frac{1}{2}, \qquad \operatorname{Var} X_r \le \mathbb{E}X_r^2 \le \frac{\mathbb{E}\xi^2}{z_r} < 1. \tag{13}$$

For `r>=2`, every value of `X_r` is at most `M_r=n_{r-1}`. The first stage can instead use `M_1=1` and has no bearing on the limit. The probabilities of zero and one in every row are at least `1/2` and `1/4`, respectively.

As `r` tends to infinity, `X_r` converges in distribution to `xi`, its mean and variance converge to `nu` and `sigma^2`, and its squares are uniformly integrable. Indeed,

$$\sup_r \mathbb{E}[X_r^2; X_r > K] \le \frac{4}{3}\mathbb{E}[\xi^2; \xi > K] \longrightarrow 0. \tag{14}$$

These are exactly the hypotheses of the conditional local CLT in agent 001, restated with its proof in Section 8 below. In particular, the limiting covariance is

$$\mathrm{Cov}(\mathbf{1}_{\{\xi=1\}}, \xi) = q(1-\nu) = \gamma, \tag{15}$$

not the covariance obtained by treating category counts as independent.

4. Exact conditioning decomposition and current rare branches

Let `xi_1,...,xi_n` be independent with law `pi`, and condition on

$$\mathcal{C}_r = \{\xi_1 + \cdots + \xi_n = n-1\}.$$

No later-stage atom can appear on `C_r`, because (7) gives `L_sˆ->n_r` for every `s>r`. Two current-stage atoms cannot appear either, because `2L_rˆ->(5/4)n>n-1`. Thus `C_r` has exactly three disjoint possible types:

- all variables belong to `A_r`;
- exactly one variable equals `L_rˆ+`, and all others belong to `A_r`;
- exactly one variable equals `L_rˆ-`, and all others belong to `A_r`.

Call their joint probabilities with `C_r beta_rˆ0`, `beta_rˆ+`, and `beta_rˆ-`, respectively. For `epsilon` equal to `+1` or `-1`, write `L_rˆepsilon=H_r+epsilon sqrt(n)`. If `R_{r,k}` is the sum of `k` independent copies of `X_r`, then exactly

$$\beta_r^\epsilon = np_r z_r^{n-1}\mathbb{P}(R_{r,n-1} = k_r^\epsilon), \qquad k_r^\epsilon = n - 1 - L_r^\epsilon. \tag{16}$$

The target and centering error have the exact expression

$$\begin{aligned} k_r^\epsilon - (n-1)\mu_r &= u_r n - 1 - \epsilon\sqrt{n} - (n-1)\frac{u_r}{z_r} \\ &= -\epsilon\sqrt{n} - d_{r-1} - (n-1)\frac{u_r t_r}{z_r} \\ &= -\epsilon\sqrt{n} - d_{r-1} + O(n^{-3}). \end{aligned} \tag{17}$$

Thus this quantity divided by `sqrt(n-1)` tends to `-epsilon`. The local CLT gives, for both signs,

$$\mathbb{P}(R_{r,n-1} = k_r^\epsilon) \sim \frac{C_\sigma}{\sqrt{n}}, \qquad C_\sigma = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-1/(2\sigma^2)} > 0. \tag{18}$$

Since `(n-1)t_r=O(nˆ{-3})`, we have `z_rˆ{n-1}=1+O(nˆ{-3})`. Consequently,

$$\beta_r^+ = C_\sigma n^{-7/2}(1 + o(1)), \qquad \beta_r^- = C_\sigma n^{-7/2}(1 + o(1)). \tag{19}$$

Equal limiting weights follow from the equal Gaussian density at the two opposite targets. No exact finite-size symmetry, independence of category counts, or unequal-probability correction device is assumed.

5. A whole-sum Bernstein estimate eliminates the remaining type

The no-current-rare probability is exactly

$$\beta_r^0 = z_r^n \mathbb{P}(R_{r,n} = n-1). \tag{20}$$

Put `D_r=n-1-n mu_r`. By (13), for every stage,

$$\frac{n}{3} \le D_r \le n. \tag{21}$$

For `r>=2`, centered summands `X_r-mu_r` are bounded above by `M_r=n_{r-1}` and have variance below one. Bernstein's inequality, applied once to their entire sum, gives

$$\begin{aligned} \mathbb{P}(R_{r,n} = n-1) &\le \mathbb{P}(R_{r,n} - n\mu_r \ge D_r) \\ &\le \exp\left[-\frac{D_r^2}{2(n \operatorname{Var} X_r + M_r D_r/3)}\right] \\ &\le \exp\left[-\frac{n}{24 M_r}\right] = \exp\left[-\frac{n^{3/4}}{24}\right]. \end{aligned} \tag{22}$$

For the final numerical constant, use

$$D_r^2 \ge n^2/9, \qquad 2(n \operatorname{Var} X_r + M_r D_r/3) \le 2n(1 + M_r/3) \le 8nM_r/3.$$

This inequality can be derived directly: if a centered variable `Z` has `|Z|<=M`, then for `0<lambda<3/M`, the exponential series and `k!>=2*3^(k-2)` for `k>=2` give

$$\log \mathbb{E} e^{\lambda Z} \le \frac{\lambda^2 \mathbb{E} Z^2}{2(1 - \lambda M/3)}.$$

Apply this to the independent centered summands and choose `lambda=D/(V+MD/3)`, where `V` is their total variance. This yields the middle line of (22). Here `|X_r-mu_r|<=M_r` because `0<=X_r<=M_r`. Thus no unverified tail theorem is essential to this step.

Equations (19), (20), and (22) imply

$$\frac{\beta_r^0}{\beta_r^+ + \beta_r^-} = O\Big(n^{7/2} e^{-n^{3/4}/24}\Big) \longrightarrow 0. \tag{23}$$

In particular,

$$\mathbb{P}(\mathcal{C}_r) \sim 2C_\sigma n^{-7/2}, \qquad \mathbb{P}(J_r = 1 \mid \mathcal{C}_r) \to 1, \qquad \mathbb{P}(\text{current sign} = \epsilon \mid \mathcal{C}_r) \to \frac{1}{2}, \tag{24}$$

where `J_r` counts current rare atoms. This proves the exact conditioning-dominance obligation left open in agent 001. It avoids the original memo's nonuniform union bound over earlier stages entirely.

6. Conditional degree-one count and the non-Gaussian limit

Given a current sign and the conditioning, the other `n-1` variables have the law of independent `X_r` variables conditioned on sum `k_r^epsilon`. The current atom is not one, so the total number of ones is just the number `B_r` of ones in those `n-1` variables.

Apply the conditional CLT using (17). It gives

$$\mathcal{L}\left(\frac{B_r-(n-1)q_r}{\sqrt{n-1}}\;\middle|\;R_{r,n-1}=k_r^{\epsilon}\right)\Longrightarrow\mathcal{N}(-\epsilon a_*,v_*). \tag{25}$$

The requested centering by `nq` introduces no hidden order-`sqrt(n)` error, because exactly

$$(n-1)q_r-nq=-q+(n-1)\frac{qt_r}{z_r}=-q+O(n^{-3}). \tag{26}$$

Also `sqrt((n-1)/n)->1`. Thus (25) remains valid with `(B_r-nq)/sqrt(n)`. Combine this with (24), testing against any bounded continuous function; the discarded no-current-rare type has conditional probability tending to zero. This proves the mixture limit for the conditioned independent sample.

For completeness, `v_*>0`, not merely nonnegative. Equality in the covariance Cauchy-Schwarz inequality would require `1_{xi=1}-q` to be an affine multiple of `xi-nu` almost surely. The left side has the same value at `xi=0` and at `xi=L_1^->1`, so the multiplier would be zero; its different value at `xi=1` then gives a contradiction. All three atoms have positive probability.

The limit's characteristic function is

$$e^{-v_*s^2/2}\cos(a_*s). \tag{27}$$

It vanishes at `s=pi/(2a_*)`, while every Gaussian characteristic function is everywhere nonzero. Thus the limit is non-Gaussian. Its symmetry and continuity also imply

$$\mathbb{P}\left((N_1(T_{n_r})-n_rq)/\sqrt{n_r}>0\right)\to\frac{1}{2}.$$

7. Transfer from conditioned allocations to conditioned trees

The count transfer is exact. A nonnegative integer vector `(x_1,...,x_n)` with sum `n-1` has increments `a_i=x_i-1` summing to `-1`. Exactly one cyclic rotation has all partial increment sums nonnegative before the final step, and this rotation encodes a plane tree's depth-first outdegrees.

Here is a short deterministic verification of uniqueness. Write $H_0=0$ and $H_j=\sum_{i\le j}a_i$, and choose the first index j minimizing $H_0,\dots,H_{n-1}$. Starting after j, the non-wrapping partial sums before the original endpoint are nonnegative because later heights are at least H_j. If the original endpoint occurs before the final rotated step, then $j>0$; the first-minimum property and integrality give $H_j\le -1=H_n$, so that partial sum is also nonnegative. Wrapped partial sums are $-1-H_j+H_\ell$ for $\ell<j$, which are nonnegative because those earlier heights are strictly larger integers. Conversely, these inequalities force the starting index to be that first minimum. No vector here has a nontrivial cyclic period, since repetitions would make its total increment sum divisible by their number, whereas that sum is -1.

The conditional independent-sample probability is proportional to `product_i pi_{x_i}` and is invariant under rotation. Each rotation orbit has `n` elements, exactly one encoding a tree, all with the same product weight. Consequently the selected valid rotation has the conditioned Galton-Watson tree law, and every rotation-invariant statistic, including degree counts and the sorted degree sequence, has exactly the same distribution before and after the rotation. This proves the transfer used in (2).

Every positive size is possible: the path with `n` vertices has probability `pi_1^(n-1) pi_0>0`. There is thus no omitted admissibility subsequence. Our subsequence is chosen for the asymptotic phenomenon, not to avoid undefined conditionings.

On the event with one current rare atom, the largest degree is `L_r^epsilon`, with

$$L_r^{\epsilon}/n_r = d_{r-1} + \epsilon n_r^{-1/2} \to 1 - \nu,$$

and the second-largest is at most `n_{r-1}=n_r^{1/4}`. Together with (24), this proves the stated unique macroscopic degree conclusion along `n_r` only.

8. Conditional local CLT dependency, reproduced for completeness

This is the lemma proved in agent 001, with notation adapted to the preceding proof. Let integer-valued `X_r` converge in distribution to `X`, with uniformly integrable squares, and suppose their masses at zero and one are bounded below by a common positive constant. Let `N_r->infinity`, `mu_r=E X_r`, `q_r=P(X_r=1)`, `mu=E X`, `q=P(X=1)`, `sigma^2=Var X>0`, `b=q(1-q)`, and `gamma=q(1-mu)`. For sums `S_r` and counts of ones `B_r` from `N_r` independent row variables, if

$$\frac{k_r - N_r\mu_r}{\sqrt{N_r}} \to a,$$

then

$$\sqrt{N_r}\,\mathbb{P}(S_r = k_r) \to \frac{e^{-a^2/(2\sigma^2)}}{\sqrt{2\pi\sigma^2}}, \tag{28}$$

and the conditional law of `(B_r-N_r q_r)/sqrt(N_r)` converges to

$$\mathcal{N}\left(\frac{\gamma a}{\sigma^2}, b - \frac{\gamma^2}{\sigma^2}\right). \tag{29}$$

To prove it, set `Y_r=1_{X_r=1}`. Distributional convergence and uniform integrability give convergence of the covariance matrix of `(X_r,Y_r)` to entries `sigma^2,gamma,b`. For fixed real `s`, Fourier inversion in the sum variable gives

$$\mathbb{E}\left[e^{is(B_r - N_r q_r)/\sqrt{N_r}}; S_r = k_r\right] = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-it(k_r - N_r\mu_r)}\left(\mathbb{E}e^{it(X_r-\mu_r)+is(Y_r-q_r)/\sqrt{N_r}}\right)^{N_r} dt. \tag{30}$$

After putting `t=v/sqrt(N_r)`, the quadratic characteristic-function expansion converges pointwise to

$$e^{-iav}\exp[-(\sigma^2 v^2 + 2\gamma v s + b s^2)/2]. \tag{31}$$

Uniform integrability of the squared centered variables justifies the expansion with an `o(1/N_r)` remainder. Explicitly, one truncates the centered linear combination at a fixed threshold, uses the Taylor remainder there, and then bounds the remaining remainder by its second-moment tail before increasing that threshold.

To justify integration, put `phi_r(t,s)=E exp(itX_r+isY_r/sqrt(N_r))`. An independent-copy expansion, retaining only the ordered pairs of atoms zero and one, gives

$$1-|\phi_r(t,s)|^2 \geq 2\mathbb{P}(X_r=0)\mathbb{P}(X_r=1)[1-\cos(t+s/\sqrt{N_r})]. \tag{32}$$

On a fixed small interval $|t|\leq\delta$, its N_r-th-power modulus is at most

$$\exp\left[-cN_r\left(t+\frac{s}{\sqrt{N_r}}\right)^2\right].$$

Under the substitution this is the integrable bound $\exp[-c(v+s)^2]$. On $\delta\leq|t|\leq\pi$ it is exponentially small uniformly in r, for each fixed s. Thus dominated convergence applied to (30) yields the Gaussian integral of (31), divided by 2π, after multiplication by $\sqrt{N_r}$.

At `s=0` this is (28). Dividing the general integral by its value at zero gives

$$\exp\left(i\frac{\gamma a}{\sigma^2}s-\frac{1}{2}\Big(b-\frac{\gamma^2}{\sigma^2}\Big)s^2\right),$$

proving (29) by characteristic functions. Nonnegativity of the variance follows from Cauchy-Schwarz. This derivation never factors the count of ones from the contribution of other categories.

The audit also checked a lattice detail. All rare degrees are multiples of 16: the recursion for H_r, the earlier pair sums, and $\sqrt{n_r}$ preserve that divisibility. Since $\gcd(176,208)=16$, the joint pair $(\xi,\mathbf{1}_{\{\xi=1\}})$ has difference-lattice index 16, although ξ itself has span one. Under the sum conditioning, the count is congruent to n_r-1 modulo 16. Its scaled spacing tends to zero, and the proof above inverts only the offspring-sum coordinate. It therefore does not require an incorrect joint aperiodicity assumption.

9. Reproducible arithmetic certificate

Script: `computations/015_lacunary_certificate.py`. Executed with Python in this session; exit code zero. It uses exact rational arithmetic for the first three stages, verifies the integrality and support bounds, and encloses the infinite tail by (5). Decimal endpoints are rounded outwards. It does not simulate conditioned trees and is not a substitute for the proof of convergence.

The first two pairs are

Stage	Conditioning size	Smaller atom	Larger atom
1	256	176	208
2	4,294,967,296	3,221,159,552	3,221,290,624

The script certifies, including the full infinite tail,

$$\nu \in [0.25000008940696716308593750001893, 0.25000008940696716308593750001894],$$
$$\sigma^2 \in [0.18751724064349328819349030655071, 0.18751724064349328819349030655072],$$
$$a_* \in [0.99990793915708316509750109871089, 0.99990793915708316509750109871090],$$
$$v_* \in [0.00001728375773097712578505130986 5713, 0.00001728375773097712578505130986 5714].$$

For example, after retaining stages through `R` and writing `m_1,m_2` for their first and second moments including the atom at one, the remaining errors are at most `4n_{R+1}^{-3}` and `4n_{R+1}^{-2}`. Hence

$$m_2 - (m_1 + 4n_{R+1}^{-3})^2 \le \sigma^2 \le m_2 + 4n_{R+1}^{-2} - m_1^2.$$

The intervals for (1) follow by monotone interval arithmetic with positive denominators. These explicit numbers illustrate that each Gaussian component is narrow and has mean close to `+1` or `-1`; the qualitative non-Gaussian conclusion needs only `a_*>0`.

10. Exact source match and limitations

Primary sources inspected through the web on 2026-09-14:

1. Paul Thévenin, *Vertices with fixed outdegrees in large Galton-Watson trees*, arXiv:1812.07365v2, Section 6.2, Theorem 6.7 and the immediately following conjecture, printed page 23. The theorem assumes a power-law mass function; its following conjecture extends the stated normal limit to subcritical offspring laws without exponential moments. For the set `{1}`, that proposed limit is `N(0,pi_1(1-pi_1))` under centering `n pi_1` and scaling `sqrt(n)`. The present law satisfies the conjecture's subcritical and exponential-moment conditions, plus the paper's nondegeneracy and span assumptions. It therefore contradicts that literal subcritical conjectural extension. It does not contradict Theorem 6.7, since this lacunary mass function is not asymptotic to a positive power law.
2. Svante Janson, *Simply generated trees, conditioned Galton-Watson trees, random allocations and condensation*, Problem 20.2, together with the canonical-law definitions and Section 8's subcritical classification. The subcritical part of Problem 20.2 is an open-ended question about degree-count fluctuations. The present result supplies a negative example to unrestricted Gaussian behavior at its displayed centering and scaling; it does not provide a classification of subcritical fluctuation laws or answer every part of that problem.

The construction starts from the project's archived Janson degree-count memo. Its full path, source snapshot, inherited gaps and repairs are documented in agent001 and agent015. Those complete reports preserve the research provenance.

The source checks establish the mathematical hypothesis match; they do not establish whether another author had already disproved or modified the conjecture by September 2026. No literature-priority or novelty clearance is claimed. The construction itself is adapted from the supplied memo. The substantive repairs are the exact law/support assembly, the uniform conditioning-dominance estimate, and application of the correct conditional covariance formula.

A critical infinite-variance offspring law with non-Gaussian degree counts

This note gives a complete counterexample to the critical infinite-variance Gaussian extension in Janson Problem 20.2 and Thévenin's closing conjecture. It integrates the construction in agent028, the complete independent audit and computable selector in agent033, and the parent proof review. Historical novelty is not certified.

The offspring law is fixed before the size of the conditioned tree varies. Its definition uses a terminating finite-stage selector. No critical-stage integers were computed; the theorem is an analytic existence and computability result, not a numerical experiment.

1. The precise theorem

There is a probability distribution π on the nonnegative integers such that

$$\pi_1 = \frac{1}{4}, \quad \pi_2 = \frac{1}{8}, \quad \pi_0 > \frac{1}{2}, \quad \mathbb{E}\xi = 1, \quad \mathbb{E}\xi^\alpha = \infty \quad \text{for every } \alpha > 1, \tag{2}$$

where $\xi \sim \pi$. In particular its variance is infinite, and it has no positive exponential moment. Its support has span one. Let T_n be its Galton-Watson tree conditioned to have n vertices. This conditioning is defined for every $n \geq 1$. For a deterministic sequence $n_r \to \infty$,

$$\frac{N_1(T_{n_r}) - n_r/4}{\sqrt{n_r}} \Longrightarrow \frac{1}{2}\mathcal{N}(-1, 3/16) + \frac{1}{2}\mathcal{N}(1, 3/16). \tag{3}$$

In particular, unrestricted convergence to $\mathcal{N}(0, 3/16)$ is false for this law. The two components have positive variance. Their mixture is not any Gaussian distribution.

2. A fixed finite-law conditional local CLT

This is the constant-row special case of agent 001's proved lemma. A proof is included to make the analytical dependency explicit.

Let Q be a finitely supported distribution on $\mathbb{Z}_{\geq 0}$, with positive masses at 0, 1, and 2. Put

$$q = Q(1), \quad \mu = \mathbb{E}_Q X, \quad V = \operatorname{Var}_Q X > 0, \quad \gamma = q(1-\mu), \quad b = q(1-q), \quad v = b - \gamma^2/V > 0. \tag{4}$$

For independent Q-distributed variables, let $S_N = \sum X_i$ and $B_N = \sum \mathbf{1}_{\{X_i=1\}}$. If integer k_N satisfy

$$\frac{k_N - N\mu}{\sqrt{N}} \longrightarrow a \in \mathbb{R},$$

then

$$\sqrt{N}\,\mathbb{P}_Q(S_N = k_N) \longrightarrow \frac{1}{\sqrt{2\pi V}} e^{-a^2/(2V)}, \tag{5}$$

and

$$\mathcal{L}_Q\left(\frac{B_N - Nq}{\sqrt{N}} \,\middle|\, S_N = k_N\right) \Longrightarrow \mathcal{N}\left(\frac{\gamma a}{V}, v\right). \tag{6}$$

Convergence in (6) is also in Kolmogorov distance, because its limiting distribution function is continuous.

To prove the lemma, set $Y = \mathbf{1}_{\{X=1\}}$. The covariance is exactly

$$\mathrm{Cov}(Y, X) = q - q\mu = \gamma. \tag{7}$$

For fixed real s, Fourier inversion yields

$$\begin{aligned} &\mathbb{E}_Q\left[e^{is(B_N - Nq)/\sqrt{N}}; S_N = k_N\right] \\ &\quad = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-it(k_N - N\mu)}\left(\mathbb{E}_Q e^{it(X-\mu)+is(Y-q)/\sqrt{N}}\right)^N dt. \end{aligned} \tag{8}$$

After substituting $t = u/\sqrt{N}$, the integrand converges pointwise to

$$e^{-iau}\exp\{-\tfrac{1}{2}(Vu^2 + 2\gamma us + bs^2)\}. \tag{9}$$

This is the ordinary quadratic characteristic-function expansion for a fixed bounded random vector. For domination, if $\phi_N(t, s) = \mathbb{E}_Q e^{itX + isY/\sqrt{N}}$, the independent-copy identity, keeping just the atom pairs 0 and 1, gives

$$1 - |\phi_N(t, s)|^2 \geq 2Q(0)q\,[1 - \cos(t + s/\sqrt{N})]. \tag{10}$$

For $|t|$ in a fixed small neighborhood of zero this bounds $|\phi_N|^N$ by $e^{-cN(t+s/\sqrt{N})^2}$; after substitution the bound is the integrable function $e^{-c(u+s)^2}$. Outside that neighborhood the integrand is exponentially small uniformly in N, for fixed s. Dominated convergence therefore gives the Gaussian integral of (9) after multiplication of (8) by $\sqrt{N}$. At $s = 0$ this is (5). Dividing the general integral by its value at zero gives

$$\exp\left\{i\frac{\gamma a}{V}s - \frac{1}{2}\left(b - \frac{\gamma^2}{V}\right)s^2\right\},$$

which proves (6). Strict positivity of v follows from strict Cauchy-Schwarz: $Y - q$ cannot be an affine multiple of $X - \mu$, because its values agree at 0 and 2 but differ at 1.

Uniform convergence of the distribution functions follows directly from weak convergence and continuity of the Gaussian distribution function. One proof is to choose finitely many partition points making all Gaussian increments and both tails small, then use convergence at those points and monotonicity between them.

The order of limits matters. This lemma will be applied with the entire finite law held fixed while choosing a size at one stage. It is not asserted uniformly for all the eventual stages. The quantitative diagonal tests below provide exactly the uniformity needed along the selected sequence.

3. The recursive stage data and all selection tests

Start with the finite probability law

$$Q^{(0)}(0) = \frac{5}{8}, \qquad Q^{(0)}(1) = q = \frac{1}{4}, \qquad Q^{(0)}(2) = \frac{1}{8}. \tag{11}$$

Its mean is $1/2$, variance is $1/2$, and the conditional variance in (4) is $5/32$. Put $n_0 = 1$. At stage $r \geq 1$, the law $Q^{(r-1)}$ is already fixed and has mean

$$\mu_r = 1 - \Delta_r, \qquad \Delta_r = 2^{-r}. \tag{12}$$

Define from this finite law

$$\begin{aligned} M_r &= \max \operatorname{supp} Q^{(r-1)}, \\ V_r &= \operatorname{Var}_{Q^{(r-1)}} X, \\ v_r &= q(1-q) - \frac{q^2\Delta_r^2}{V_r} > 0, \\ \varepsilon_r &= \frac{1}{r+10}, \\ c_r &= \frac{1}{\sqrt{2\pi V_r}} \exp\left(-\frac{V_r}{2q^2\Delta_r^2}\right) > 0. \end{aligned} \tag{13}$$

For each candidate $n = 2^\ell$, let $N = n - 1$ and define

$$D_r(n) = \frac{V_r\sqrt{N}}{q\Delta_r}, \qquad L_r^{\pm}(n) = \lfloor N\Delta_r \pm D_r(n) \rfloor, \qquad k_r^{\pm}(n) = N - L_r^{\pm}(n). \tag{14}$$

Here all support points are integers; irrationality of $D_r(n)$ creates no mean-adjustment problem.

Choose n_r by the deterministic certificate enumeration in Section 10, accepting a power of two only when all the following tests hold. Every law in the tests is the already fixed law $Q^{(r-1)}$. The selection does not require deciding whether a computable real number is exactly zero.

G1: growth and support separation.

$$n \geq 2^{8r}, \qquad n \geq 2n_{r-1}, \qquad M_r \leq \varepsilon_r n\Delta_r, \qquad D_r(n) + \Delta_r + 1 \leq \varepsilon_r n\Delta_r, \qquad L_r^-(n) < L_r^+(n). \tag{15}$$

G2: both local probabilities have a prescribed relative error. For each sign,

$$\left| \frac{\sqrt{N}\,\mathbb{P}_{Q^{(r-1)}}(S_N = k_r^{\pm}(n))}{c_r} - 1 \right| < \varepsilon_r. \tag{16}$$

G3: both conditional count laws have a prescribed error. Writing d_K for Kolmogorov distance,

$$d_K\left(\mathcal{L}_{Q^{(r-1)}}\left(\frac{B_N - Nq}{\sqrt{N}} \,\middle|\, S_N = k_r^{\pm}(n)\right), \mathcal{N}(\mp 1, v_r)\right) < \varepsilon_r. \tag{17}$$

G4: the no-current-pair route is negligible.

$$\exp\left\{-\frac{(n\Delta_r - 1)^2}{2(nV_r + M_r(n\Delta_r - 1)/3)}\right\} < \varepsilon_r \, \frac{c_r n^{-3}}{\sqrt{N}}. \tag{18}$$

G5: all routes with at least two current-pair atoms are negligible.

$$2n^{-6} < \varepsilon_r \, \frac{c_r n^{-3}}{\sqrt{N}}. \tag{19}$$

Together with the selector in Section 10, these gates specify the stage. Tests G2 and G3 are finite probability calculations. More explicitly, if

$$F_r(z,u) = Q^{(r-1)}(0) + qzu + \sum_{j \geq 2} Q^{(r-1)}(j) z^j, \tag{20}$$

then

$$\mathbb{P}_{Q^{(r-1)}}(S_N = k, B_N = b) = [z^k u^b] F_r(z,u)^N. \tag{21}$$

All its coefficients are rational. For (17), the conditional distribution has finitely many jumps, at $(b - Nq)/\sqrt{N}$; the supremum is the maximum of the differences from the Gaussian CDF at the left and right sides of those jumps. Therefore (17) also uses a finite list of explicitly defined numbers.

Section 10 fixes an enumeration of candidate and precision pairs and selects the first candidate whose strict gates have positive rational lower bounds. Every proof below uses only G1-G5. Existence of a passing candidate, proved next, also proves termination of that selector. No stage integer has been computed in this run, and no numerical efficiency is asserted.

Every stage's selection set is nonempty

Hold r and $Q^{(r-1)}$ fixed and let candidate powers of two tend to infinity. All terms $M_r, V_r, \Delta_r, c_r, v_r, \varepsilon_r$ are fixed finite constants, with the needed ones positive. G1 eventually holds, because $D_r(n) = O(\sqrt{n})$, $n\Delta_r$ is linear, and the earlier support is fixed. Also

$$\frac{k_r^{\pm}(n) - N\mu_r}{\sqrt{N}} = \mp \frac{V_r}{q\Delta_r} + O(N^{-1/2}). \tag{22}$$

The finite-law lemma gives (16), since (5)'s Gaussian factor is exactly c_r. Equation (6) gives the target mean

$$\frac{q\Delta_r}{V_r}\left(\mp \frac{V_r}{q\Delta_r}\right) = \mp 1, \tag{23}$$

and the variance is exactly v_r, so (17) eventually holds. The exponent on the left of (18) grows linearly in n, while its right side is a positive fixed constant times $n^{-7/2}(1+o(1))$. Thus (18) eventually holds. Finally, the ratio of the left of (19) to its right side is

$$\frac{2\sqrt{n-1}}{\varepsilon_r c_r n^3} = O_r(n^{-5/2}) \longrightarrow 0. \tag{24}$$

Hence (19) eventually holds as well. There are finitely many tests, so sufficiently large powers of two pass all of them. This proves nonemptiness and completes the diagonal-existence argument.

4. The exact probability and mean update

At the selected size write $L_r^{\pm} = L_r^{\pm}(n_r)$, and set

$$p_r = n_r^{-4}, \qquad H_r = n_r^4, \qquad a_r = \frac{\Delta_r/2 - p_r(L_r^- + L_r^+)}{H_r}. \tag{25}$$

Add mass p_r at each of L_r^- and L_r^+, add mass a_r at the integer H_r, and subtract total mass

$$h_r = 2p_r + a_r \tag{26}$$

from zero. Call the resulting finite law $Q^{(r)}$. All its probabilities are rational. No already assigned positive support mass changes at later stages.

From G1, since $\varepsilon_r < 1/10$ and $\Delta_r \le 1/2$,

$$M_r < L_r^- < L_r^+ < n_r < H_r, \qquad (1-\varepsilon_r)n_r\Delta_r \le L_r^- \le L_r^+ \le (1+\varepsilon_r)n_r\Delta_r. \tag{27}$$

For the first strict inequality, use $M_r \le \varepsilon_r n_r \Delta_r$ and $1-\varepsilon_r > \varepsilon_r$. The upper bound follows from $L_r^+ \le (n_r-1)\Delta_r + D_r$. Thus every stage has three fresh, disjoint support points, none equal to 0, 1, or 2. At the next stage the actual background maximum is

$$M_{r+1} = H_r = n_r^4. \tag{28}$$

In particular $n_{r+1} > n_r^4$. The adjustment atom's full size is carried into the next stage, rather than incorrectly being bounded by n_r.

Since both current pair atoms are below n_r,

$$0 < p_r(L_r^- + L_r^+) < 2n_r^{-3} \le \Delta_r/4. \tag{29}$$

The final inequality follows from $n_r \ge 2^{8r}$. Consequently

$$\frac{\Delta_r}{4H_r} < a_r < \frac{\Delta_r}{2H_r}, \qquad 0 < h_r < 3n_r^{-4}. \tag{30}$$

The mean increment is *exactly*

$$p_r(L_r^- + L_r^+) + a_r H_r = \Delta_r/2. \tag{31}$$

Therefore the updated mean is $1 - \Delta_r/2 = 1 - \Delta_{r+1}$, proving the induction used in (12). Floors in (14) cause no problem because the adjustment uses a real probability at a fixed integer atom, rather than asking an integer atom size to match a prescribed real mean with fixed probability.

Because $n_{s+1} \geq 2n_s$,

$$\sum_{s \geq r} h_s \leq 3 \sum_{s \geq r} n_s^{-4} \leq \frac{3n_r^{-4}}{1 - 2^{-4}} < 4n_r^{-4}. \tag{32}$$

In particular the total mass subtracted from zero is less than $4 \cdot 256^{-4} < 1/8$. Every finite update remains a probability distribution, and its zero mass remains above $1/2$.

Define the single final law by

$$\pi_1 = \frac{1}{4}, \quad \pi_2 = \frac{1}{8}, \quad \pi_{L_r^-} = \pi_{L_r^+} = p_r, \quad \pi_{H_r} = a_r, \quad \pi_0 = \frac{5}{8} - \sum_{r \geq 1} h_r, \tag{33}$$

and give every other integer mass zero. Equations (25)-(33), together with the finite selection rule, define one fixed law, not an n-dependent collection of offspring laws.

5. Exact criticality and moment properties

The base mean is $1/2$. Summing (31) and using nonnegativity gives

$$\mathbb{E}_\pi \xi = \frac{1}{2} + \sum_{r \geq 1} \frac{\Delta_r}{2} = \frac{1}{2} + \sum_{r \geq 1} 2^{-r-1} = 1. \tag{34}$$

Thus the law is exactly critical. For every $\alpha > 1$, (30) gives a single-term lower bound

$$a_r H_r^\alpha > \frac{2^{-r}}{4} n_r^{4(\alpha-1)}. \tag{35}$$

The support separation gives $n_{r+1} > n_r^4$, so $\log n_r \geq 4^{r-1} \log 256$. The logarithm of the right side of (35) tends to infinity for every $\alpha > 1$. Hence $\mathbb{E}\xi^\alpha = \infty$ for every such α. In particular $\mathbb{E}\xi^2 = \infty$. For $0 < \alpha \leq 1$, finiteness follows from $x^\alpha \leq 1 + x$.

The second moments of $Q^{(r)}$ increase to infinity. Their means are bounded by one, so

$$V_r \longrightarrow \infty, \qquad v_r \longrightarrow q(1-q) = 3/16. \tag{36}$$

The law has no positive exponential moment; otherwise all its positive moments would be finite. Equivalently, one can use the adjustment terms directly: $a_r e^{\theta H_r} \geq 2^{-r} e^{\theta n_r^4}/(4n_r^4) \to \infty$ for every $\theta > 0$. Therefore its probability generating function has radius exactly one. For simply generated weights $w_k = \pi_k$, the canonical law is this same π, with $\rho = \tau = 1$, $\Phi(1) = 1$, and $\Psi(1) = 1$. Positivity at 0 and 1 gives lattice span one.

6. Future mass does not change the rare conditional laws

Fix stage r, and abbreviate $n = n_r$, $N = n - 1$, $Q = Q^{(r-1)}$, and $A = \operatorname{supp} Q$. Let

$$\tau_r = \sum_{s \geq r} h_s < 4n^{-4}, \qquad z_r = Q(0) > 1/2. \tag{37}$$

On positive points of A, the final law π equals Q exactly. At zero it equals $Q(0) - \tau_r$. Thus for any vector $x \in A^k$, $k \leq n$,

$$\prod_{i=1}^{k} \pi_{x_i} = \prod_{i=1}^{k} Q(x_i) \left(1 - \frac{\tau_r}{z_r}\right)^{\#\{i : x_i = 0\}}. \tag{38}$$

Consequently every such product lies between $\eta_r \prod Q(x_i)$ and $\prod Q(x_i)$, where

$$\eta_r := \left(1 - \frac{\tau_r}{z_r}\right)^n \geq 1 - \frac{n\tau_r}{z_r} \geq 1 - 8n^{-3} \longrightarrow 1. \tag{39}$$

This is a **multiplicative** bound valid for every configuration and every event on the background coordinates, however small its probability. An additive total variation estimate of order $n\tau_r$ alone would not suffice for an exponentially small conditional denominator. Equation (38) avoids that issue completely.

After conditioning on any positive-probability background event, the reweighted law has a density with respect to the corresponding conditioned Q law between η_r and η_r^{-1}. Its total variation distance is therefore at most

$$\eta_r^{-1} - 1 = O(n^{-3}). \tag{40}$$

This handles the recursive dependence on the final zero probability; no exact knowledge of future support or probabilities was needed in the finite-stage CLT tests.

7. Conditioning decomposition, both branch weights, and every other rare count

Take independent variables $\xi_1, \ldots, \xi_n$ with the final law π, and put

$$\mathcal{C}_r = \left\{ \sum_{i=1}^{n} \xi_i = n - 1 \right\}. \tag{41}$$

The current adjustment atom is $H_r = n_r^4 > n_r$. Every future support point exceeds H_r, by (27)-(28) and induction. Hence no current adjustment atom and no future atom can occur on $\mathcal{C}_r$, exactly. Earlier adjustment atoms are allowed and are included in Q, with maximum M_r.

Let J_r be the number of current-pair atoms, namely coordinates equal to L_r^- or L_r^+. On $\mathcal{C}_r$, every coordinate is either in the background A or is a current-pair atom. The possibilities $J_r = 0, 1, 2, \ldots$ exhaust the conditioning event. Unlike the subcritical construction, multiple current-pair atoms are not declared impossible.

Write $\beta_r^{\pm}$ for the joint probability of $\mathcal{C}_r$ and exactly one current-pair atom, with sign $\pm$. Put

$$w_r := np_r\frac{c_r}{\sqrt{N}} = \frac{c_r n^{-3}}{\sqrt{N}} > 0. \tag{42}$$

By (38),

$$\eta_r np_r \mathbb{P}_Q(S_N = k_r^{\pm}) \le \beta_r^{\pm} \le np_r \mathbb{P}_Q(S_N = k_r^{\pm}). \tag{43}$$

The factor n counts the position of the single current-pair atom. Using G2,

$$\eta_r(1-\varepsilon_r)w_r \le \beta_r^{\pm} \le (1+\varepsilon_r)w_r. \tag{44}$$

Thus both branch probabilities are asymptotic to the **same** changing quantity w_r. The two Gaussian local densities agree at opposite targets, and the finite selection tests make that symmetry accurate enough; no unproved exact symmetry or recursive unequal-probability adjustment is used.

For the route $J_r = 0$, (38) gives

$$\begin{aligned}\mathbb{P}(\mathcal{C}_r, J_r = 0) &\le \mathbb{P}_Q(S_n = n-1) \\ &\le \exp\left\{-\frac{(n\Delta_r - 1)^2}{2(nV_r + M_r(n\Delta_r - 1)/3)}\right\} < \varepsilon_r w_r.\end{aligned} \tag{45}$$

The middle inequality is the usual Bernstein bound for independent bounded centered summands. Here $0 \le X \le M_r$, $|X - \mu_r| \le M_r$, variance is exactly V_r, and the required excess is $n\Delta_r - 1 > 0$. A direct derivation is to use

$$\log \mathbb{E}e^{t(X-\mu_r)} \le \frac{t^2 V_r}{2(1 - tM_r/3)}, \qquad 0 < t < 3/M_r,$$

which follows by expanding the exponential and using $k! \ge 2 \cdot 3^{k-2}$ for $k \ge 2$, then optimize with $t = D/(nV_r + M_r D/3)$, where $D = n\Delta_r - 1$. This proves the displayed bound without another local-limit input. The last inequality is G4.

For **all** routes $J_r \ge 2$, use the unconditional factorial moment of the current-pair count. In the independent final sample, $J_r \sim \mathrm{Bin}(n, 2p_r)$. Therefore

$$\begin{aligned}\mathbb{P}(\mathcal{C}_r, J_r \ge 2) \le \mathbb{P}(J_r \ge 2) &\le \frac{1}{2}\mathbb{E}[J_r(J_r - 1)] \\ &= \binom{n}{2}(2p_r)^2 \le 2n^2 p_r^2 = 2n^{-6} < \varepsilon_r w_r.\end{aligned} \tag{46}$$

The last inequality is G5. This bounds the sum of *every* possible current-pair count at least two in one step. It does not assume that the number of allowable rare atoms is uniformly bounded, and it does not factor category counts.

Combining (44)-(46) gives

$$\mathbb{P}(\mathcal{C}_r) = 2w_r(1 + o(1)), \quad \mathbb{P}(J_r = 1 \mid \mathcal{C}_r) \longrightarrow 1, \quad \mathbb{P}(\text{single sign} = \pm \mid \mathcal{C}_r) \longrightarrow \frac{1}{2}. \tag{47}$$

For example, the total conditional probability of the bad routes is at most $\varepsilon_r/[\eta_r(1-\varepsilon_r)]$, which tends to zero. Even if c_r tends to zero extraordinarily quickly, (18)-(19) compare both errors directly with w_r. This is the denominator comparison absent from the archival critical sketch.

8. Conditional count limits and centering

On a single-sign branch the other $N = n - 1$ coordinates lie in A, and their sum is $k_r^{\pm}$. The current atom is not one, so the total count of ones is the background count B_N.

Under Q, G3 gives Kolmogorov distance less than ε_r from $\mathcal{N}(\mp 1, v_r)$ for $(B_N - Nq)/\sqrt{N}$. Equation (40) changes this by at most $O(n^{-3})$ under the true final law and the same branch conditioning. By (36), this proves

$$\mathcal{L}\left(\frac{B_N - Nq}{\sqrt{N}} \,\middle|\, \mathcal{C}_r, \text{single sign } \pm\right) \Longrightarrow \mathcal{N}(\mp 1, 3/16). \tag{48}$$

The requested centering is nq, not Nq. The exact conversion is

$$\frac{B_N - nq}{\sqrt{n}} = \sqrt{\frac{N}{n}}\,\frac{B_N - Nq}{\sqrt{N}} - \frac{q}{\sqrt{n}}. \tag{49}$$

Both changes vanish in the limit. No normalization of the background one mass, and no hidden $\sqrt{n}$-order correction from future stages, is present; the one probability of every partial law is exactly q, and (38) handles the final restriction multiplicatively.

Combining (47) and (48)-(49) against bounded continuous test functions yields (3) for the independent sample conditioned on its sum. Its limiting characteristic function is

$$e^{-3t^2/32}\cos t. \tag{50}$$

It vanishes at $t = \pi/2$, whereas the characteristic function of every Gaussian probability distribution is nonzero everywhere. Thus the limit is non-Gaussian.

Why the vanishing critical covariance is not an obstruction

The background covariance is $q\Delta_r$, which tends to zero. The target sum displacement is chosen as

$$\frac{k_r^{\pm} - N\mu_r}{\sqrt{N}} \simeq \mp\frac{V_r}{q\Delta_r}. \tag{51}$$

Its product with the regression coefficient $q\Delta_r/V_r$ remains ∓ 1. Equation (23), not the incorrect covariance $q(1-q)$, is the reason for the separation in (14). At a heuristic tilt level the required parameter is $\mp 1/(q\Delta_r\sqrt{n})$, but no heuristic tilt calculation is needed for the proof. The diagonal procedure ensures that every finite-stage Gaussian assertion remains sufficiently accurate at that stage's very large, fixed standardized displacement.

9. Transfer to conditioned Galton-Watson trees

The transfer of degree counts is exact. A vector of nonnegative integers $(x_1, \dots, x_n)$ summing to $n-1$ has increments $x_i - 1$ summing to -1. Exactly one of its cyclic rotations has all partial sums nonnegative before the final step; that rotation is the depth-first offspring sequence of a plane rooted tree.

For completeness, let $H_0 = 0$ and $H_j = \sum_{i \le j}(x_i - 1)$. The starting point is the first minimum, indexed by j, among $H_0, \dots, H_{n-1}$. Later heights before the endpoint are at least that minimum. If the original endpoint is encountered before the final rotated step, then $j > 0$, so $H_j < 0$ and integrality gives $H_j \le -1 = H_n$. Earlier heights are strictly larger integers, making the wrapped partial sums $-1 - H_j + H_\ell$ nonnegative for $\ell < j$. Conversely these inequalities force the first minimum. A nontrivial cyclic period is impossible because the total increment sum is -1.

The conditional independent-sample weights are proportional to $\prod_i \pi_{x_i}$ and are rotation invariant. Every orbit has n elements and exactly one tree sequence. The selected tree has weight proportional to the same product, which is precisely the conditioned Galton-Watson law. Counts of fixed degrees are invariant under the rotation. Therefore (3) transfers unchanged from the conditioned sample to T_{n_r}.

The path of length $n-1$ has probability $\pi_1^{n-1}\pi_0 > 0$; all sizes are admissible. Criticality and nondegeneracy imply the unconditioned tree is finite almost surely, although only the positivity of these finite-size probabilities is needed here.

10. The terminating computable selector

Fix a stage and omit its subscripts: $Q = Q^{(r-1)}$, $\Delta = 2^{-r}$, $V = V_r$, $M = M_r$, $v = v_r$, $\varepsilon = \varepsilon_r$, and $c = c_r$. Write

$$E_r(n) = \frac{(n\Delta - 1)^2}{2(nV + M(n\Delta - 1)/3)}.$$

A literal least-passing-power rule is well-defined by Section 3, but strict comparisons of computable real numbers do not by themselves provide a terminating decision for a rejected equality case. The selector below avoids such decisions. It was supplied and independently justified in agent033.

10.1 Exact finite formula for the G3 distance

All probabilities in the fixed finite law Q are rational. With

$$F(z,u) = Q(0) + qzu + \sum_{j \ge 2} Q(j) z^j,$$

define, for each chosen target k,

$$P_{k,b} = [z^k u^b] F(z,u)^N, \quad P_k = \sum_{b=0}^{N} P_{k,b}, \quad C_b = \frac{\sum_{j=0}^{b} P_{k,j}}{P_k}, \quad C_{-1} = 0. \tag{A17}$$

These are rational and are obtained by a finite exact calculation. At a G1-passing candidate $P_k > 0$, as shown above. Let

$$x_b = (b - Nq)/\sqrt{N}, \qquad G(x) = \Phi((x-m)/\sqrt{v}), \qquad m = \mp 1.$$

Then the exact Kolmogorov distance in G3 is

$$K = \max_{0 \le b \le N} \max\{|C_b - G(x_b)|,\ |C_{b-1} - G(x_b)|\}. \tag{A18}$$

Between two consecutive possible count values the discrete CDF is constant and G is increasing, so its maximum absolute difference occurs at an endpoint. The two unbounded tails are covered by the first and last endpoints, their limits at infinity being zero. Formula (A18) remains valid if some count values have zero probability.

10.2 Uniformly computable positive margins

The finite-stage mean and variance are rational. All floor arguments in (14) are algebraic numbers; their floors and all the G1 comparisons are decidable exactly by algebraic-number arithmetic. In fact, for eligible $n = 2^\ell$ with $\ell \ge 8$, $N = 2^\ell - 1$ is not a square, since it is 3 modulo 4. The nonzero rational multiple D of $\sqrt{N}$ is irrational. No ambiguous floor equality arises.

For a G1-passing candidate define the six margins

$$g_{2,\pm} = \varepsilon - \left| \frac{\sqrt{N} P_{k^\pm}}{c} - 1 \right|, \qquad g_{3,\pm} = \varepsilon - K_\pm,$$

$$g_4 = \varepsilon c n^{-3}/\sqrt{N} - e^{-E_r(n)}, \qquad g_5 = \varepsilon c n^{-3}/\sqrt{N} - 2n^{-6}. \tag{A19}$$

These are uniformly computable real numbers from the finite stage data and candidate exponent. Rational operations, positive square roots, the exponential, and the normal CDF can all be enclosed by rational intervals of any requested error. For the last function one may use its defining Gaussian integral, Taylor approximation on a finite interval, and an elementary Gaussian tail bound; all arguments here are finite and computable. Finite maxima and absolute values preserve computability. The positive denominator c permits computable division by refinement until a positive lower bound is found.

Thus, for each candidate and precision j, a terminating finite calculation produces an enclosing rational interval of width at most 2^{-j} for each margin. A strictly positive lower endpoint is a sound certificate for that gate. Margins equal to zero never receive such a certificate, which is exactly appropriate for strict gates.

10.3 Deterministic algorithm

At stage r, let ℓ_0 be the least integer for which $2^{\ell_0} \ge \max(2^{8r}, 2n_{r-1})$. Enumerate pairs (ℓ, j), $\ell \ge \ell_0$, $j \ge 1$, by increasing $(\ell - \ell_0) + j$, breaking ties by increasing ℓ.

For each pair:

1. Check G1 exactly. If it fails, continue to the next pair.
2. Compute the finite rational probabilities (A17), the exact formula (A18), and rational enclosures of all six margins (A19) at precision j.
3. If all lower endpoints are positive, choose $n_r = 2^\ell$ and end this stage. Otherwise continue.

Each visit performs only finite work. Section 3 gives an exponent ℓ with G1 and all six strict margins positive. For that exponent, sufficiently large j certifies every margin. The specified enumeration reaches that pair after finitely many visits, so the algorithm terminates. This proves termination inductively at every stage, even though no usable running-time bound is claimed.

The selected candidate satisfies the original G1-G5. Every subsequent proof uses only those gates, not minimality of the exponent. Therefore this algorithm defines a fixed law satisfying (2)-(3). It may differ from the law selected by the literal least-passing-power definition in agent028; that distinction does not change the theorem. This note adopts this certificate selector throughout.

10.4 Computability of the resulting infinite law

Under the certificate selector, every finite list of support points and positive masses is computable exactly, and the masses are rational. The support strictly increases past each stage maximum. To determine π_k for an integer $k > 0$, generate stages until a maximum $H_R \geq k$; every later positive point exceeds H_R, so membership and mass are decided by a finite list.

The only infinite sum in a mass is at zero. Since $h_s < 3n_s^{-4}$ and $n_s \geq 2^{8s}$, its omitted tail after stage R is bounded by

$$\sum_{s>R} h_s < \frac{3\,2^{-32(R+1)}}{1-2^{-32}}. \tag{A20}$$

This is an effective tail bound tending to zero. It proves that π_0, and hence the whole law, is computable. Nothing in this argument asserts that a selected critical-stage integer has actually been computed.

11. Direct exclusion of stable-domain hypotheses

This is an independent check, not an appeal to the counterexample conclusion. Let

$$U(x) = E_\pi[\xi^2 \mathbf{1}_{\{\xi \leq x\}}], \qquad W(x) = \mathrm{Var}_\pi(\xi \mathbf{1}_{\{\xi \leq x\}}).$$

At $H = H_r = n_r^4$, the interval $(H/2, H)$ contains no support point: the current pair is below $n_r < H/2$, all earlier points are smaller still, and all future points exceed H. Therefore

$$U(H/2) = E_{Q^{(r-1)}} X^2 + p_r((L_r^-)^2 + (L_r^+)^2) \leq M_r + 2n_r^{-2} \leq \varepsilon_r n_r \Delta_r + 2n_r^{-2}. \tag{A40}$$

On the other hand, $U(H) \geq a_r H_r^2 > \Delta_r n_r^4/4$. Since all truncated first moments are at most $E_\pi \xi = 1$,

$$\frac{W(H_r)}{W(H_r/2)} \geq \frac{\Delta_r n_r^4/4 - 1}{\varepsilon_r n_r \Delta_r + 2n_r^{-2}} \longrightarrow \infty. \tag{A41}$$

The denominator is positive because the law has positive masses at 0 and 1. The divergence follows from $n_r \geq 2^{8r}$, $\Delta_r = 2^{-r}$, and $\varepsilon_r = 1/(r+10)$. If $W(x)$ were regularly varying with a finite exponent $2-\alpha$, its ratio at H_r and $H_r/2$ would converge to $2^{2-\alpha}$, a finite number. Thus this law fails the truncated-variance regular-variation condition for every $\alpha \in (1, 2]$, including the normal domain with infinite variance.

12. Original conjecture match and exact scope

Janson's original paper, Problem 20.2 on printed page 135, asks about extending its Gaussian degree-count limit to critical laws with infinite variance. Thévenin's paper, version 2 of 26 February 2020, states the corresponding unrestricted critical conjecture on printed page 23. For the degree set $\{1\}$, the proposed centered limit is $\mathcal{N}(0, 3/16)$. The fixed law above has the required criticality, infinite variance, nondegeneracy and span, and (3) contradicts that limit. The original papers' proved theorems retain further tail hypotheses; Section 11 shows directly why their stable-domain hypotheses do not hold here.

Agent033 independently retrieved both primary sources and checked the literal hypotheses. This is a negative example for the specified Gaussian extension, not a classification of every possible critical fluctuation law. No full-sequence limit or moment convergence of the count is asserted. Along the selected sizes, the dominant rare degree has size $(1 + o(1))n_r\Delta_r = o(n_r)$; the remaining degrees are at most $M_r \le \varepsilon_r n_r \Delta_r$.

The complete derivations, archival proposal, rejected approaches and audit details are retained in reports001, 028 and 033. The optional law selected by the original literal minimum need not be the same law as this note's certificate-selected version. Both satisfy G1-G5 and the theorem's conclusion. Computability here refers specifically to the selector in Section 10. No practical stage computation, proof-assistant verification, external referee acceptance, or worldwide priority is claimed.